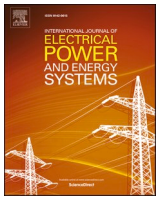
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A distributionally robust resilience enhancement model for transmission and distribution coordinated system using mobile energy storage and unmanned aerial vehicle

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ABSTRACT

Extreme natural disasters can seriously disrupt power systems. The increased damage intensity of natural disasters also leads to synchronous failures in communication systems. Mobile energy storage and unmanned aerial vehicles have high economy and flexibility, so they can provide various services including power support and temporary information transmission when disasters occur and disable the whole system. To address the unavailability of the communication system and improve the collaborative restoration of transmission and distribution systems, this paper proposes a multi-period distributionally robust resilience enhancement model (MPDRRM) for the transmission and distribution coordinated system, and a large quantity of mobile energy storage and unmanned aerial vehicles are rented before natural disasters and dispatched to support critical loads and maintain power balance during restoration. The MPDRRM is divided into three kinds of sub-problems: i) a communication system sub-problem, ii) a transmission system sub-problem, iii) several distribution system sub-problems, and solved by a modified three-level analytical target cascading algorithm. Case studies demonstrate the benefits of mobile energy storage and unmanned aerial vehicles in improving load restoration and increasing the resilience of a TDCS against natural disasters.

1. Introduction

Extreme natural disasters, such as ice disasters and earthquakes, occur frequently as the climate changes in recent decades, leading to the concern of large-scale blackouts in the power systems [1]. The severe power system failure caused by extreme natural disasters has attracted researchers to explore technologies and facilities for resilient power systems [2]. Although considerable efforts have been put into resilient power systems in recent years [3], power outages caused by extreme disasters with small probability and high loss still cannot be completely avoided [4]. In mid-September 2018, typhoon Mangkhut caused over 13,000 power outages in Taiwan, triggering damage to over 300 transmission lines [5]. From 2015 to 2022, California experienced unprecedented catastrophic wildfires [6]. Especially in 2021, the wildfire burning in Siskiyou County destroyed thousands of power loads [7]. The frequent wildfires in California also caused the Pacific Gas & Electric Company (PG&E) to file for bankruptcy in 2020 [8]. Therefore, an efficient resilience enhancement scheme, which includes the network hardening scheme in the pre-disaster stage and the restoration scheme in the post-disaster stage, is necessary for modern power systems. For such motivation, numerous papers have been proposed to improve the resilience of power systems and the resilience metric method in recent years. In [1], the authors focused on the resilience enhancement strategy under ice disasters and proposed a full-time scale resilience enhancement framework containing the whole process of extreme events. The authors in [9] used the analytic network process to establish a resilience assessment index system and its corresponding comprehensive assessment method for distribution networks. Reference [10] presented a valley-shaped resilience model to evaluate the performance of integrated power and natural gas systems under extreme weather and inter-energy assistances between the electric power sector and the natural gas sector. Also focusing on resilience metrics, the authors in [11] presented a novel multi-stage resilience measure called the Anticipate-Withstand-Recover (AWR) metrics for distribution system resilience enhancement,

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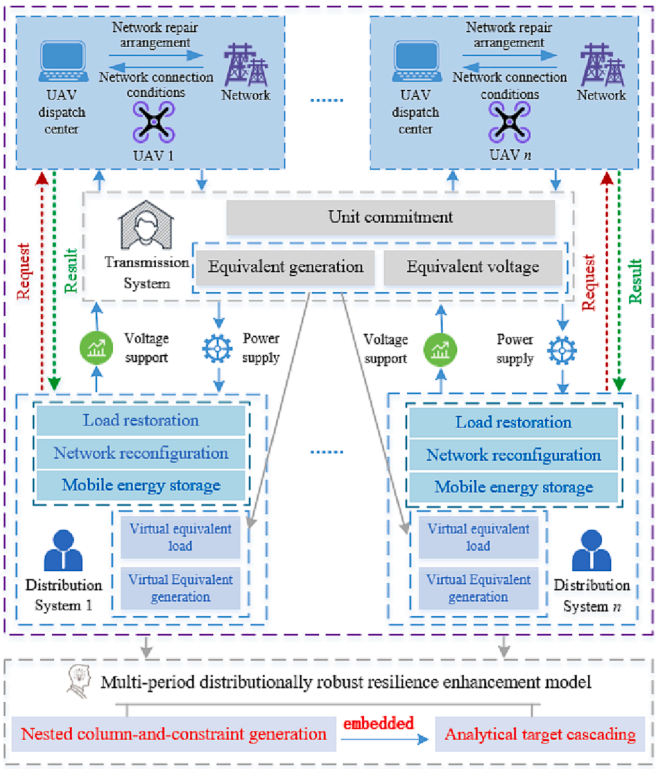


Fig. 1. The framework of the proposed MPDRRM.

and AWR metrics were based on integrating relevant 'system characteristics based factors'. Reference [12] proposed a resilience assessment method called "trapezoid" to extend the resilience "triangle" that had been traditionally used in existing studies. This method was quantified using time-dependent resilience metrics to help capture the critical system degradation for different temporal phases of an event. Reference [13] provided an introduction to the fundamental concepts of power systems resilience and the application of hardening operational strategies to improve it. The authors in [14] proposed the development of the probabilistic extreme event model, impact assessment model, and optimal restoration model for active distribution systems, and integrated these models using a non-sequential Monte Carlo Simulation framework.

Due to the gradual expansion of power systems, a transmission system (TS) along with several different distribution systems (DSs) are connected via numerous distribution substations [15], and they will be simultaneously affected by a natural disaster [16]. During natural disasters, the coordination of TS and DSs can improve voltage stability [17] against abnormal conditions, ensuring that transmission and distribution coordinated systems (TDCSs) can maintain power balance [18] and three-phase symmetric operation [19] at each period. DSs with high penetration of distributed energy resources, e.g., distributed generations and emergency power supplies, can also provide effective support for the operation and restoration of the TS via contact lines and distribution substations, which has triggered the need for coordination and interaction between transmission and distribution levels [20]. Besides, although the current power systems can predict extreme natural disasters before they occur, the uncertainties caused by sudden climate change and human factors may influence the coverage area of damage [21]. Therefore, the transmission and distribution systems have to coordinate and interact with each other during natural disasters to improve resilience [22]. For most countries, TSs and DSs are separately managed by transmission system operators (TSOs) and distribution system operators (DSOs). During the network restoration, TSs and DSs can only exchange part of the operation data, leading to difficulty in effectively implementing the collaborative recovery in TDCS [23].

To address the difficulties in collaborative operations in TDCSs or other integrated systems, various mathematical programming and algorithms have been proposed in recent years, which iteratively solved mathematical models by decomposing the original problem into several distributed sub-problems and exchanging partial boundary information through communication systems [24]. Similar algorithms include the analytical target cascading (ATC) [25], alternating direction multiplier method (ADMM) [26,27], Benders decomposition (BD) [28], heterogeneous decomposition (HGD) [29], etc. However, these algorithms have certain limitations. Ref. [30] proposed a decentralized robust model for the optimal operation of distribution companies with private microgrids (MGs) and applied the ADMM to solve it, but the ADMM algorithm cannot guarantee convergence in non-convex models. Although the HGD algorithm based on Karush-Kuhn-Tucker (KKT) conditions proposed in [29] and [31] can solve several optimal problems in a decentralized way, it can only theoretically guarantee the local convergence of the problem. The BD based on KKT conditions can certainly guarantee global convergence, but the construction of Benders cuts limits the solution efficiency [28]. The ATC presented in [16] can solve the collaborative operations in TDCS by decomposing the original problem into a master problem and several sub-problems. By coordinating and optimizing the elements in the master problem over the decisions of the elements in sub-problems, the ATC can obtain the overall optimal solution and guarantee strict convergence [32]. Based on the parallel processing of multiple sub-problems, the ATC also has high efficiency in solving the original problem [33].

Although the early works provide important insights into flexible operation measures for TDCS, they rarely consider communication system failures under natural disasters. Nearly all studies treat the communication system operating in the normal condition, in which emergency operating instructions can be delivered to every bus promptly [34]. However, communication systems are strongly coupled with physical systems. Thus, communication systems will come into a partial or complete disability when natural disasters attack power systems [35]. The communication difficulties would make the system operators unable to precisely locate the faulted transmission lines and the power loss, hence the restoration schemes in the post-disaster stage cannot be formulated and implemented promptly [36]. To this end, some new technologies and facilities, such as unmanned aerial vehicles (UAVs) and mobile energy storage (MS), can improve and recover the power grid and communication system. MS applied in power systems can achieve a flexible balance between supply and demand at any bus. Due to its flexibility and efficiency, MS can be dispatched quickly between several neighboring buses when natural disasters occur. Especially for critical nodes such as local hospitals and command centers, the fast response time and real-time voltage control allow MS to provide multiple services such as energy arbitrage, voltage regulation [37,38], and reactive power support [39], enabling these nodes to maintain voltage stability and continuous power supply. However, few studies have been applied to MS for power system resilience enhancement these years. As for UAVs, by acting as temporary base stations in the air, multiple UAVs can form a temporary communication network and replace the disabled conventional communication network. For instance, the fixed-wing UAV has a maximum flight speed of 370 km/h and an endurance of over 24 h. In 2021, several UAVs with mobile base stations were urgently deployed to Gongyi, Henan Province to restore communication for over 5 h in heavy rainfall conditions [39]. However, limited by the communication capacity of UAVs, it is impossible to transmit information about the whole system through the UAV-made communication network, especially for a TDCS. Thus, a proper model and dispatch method for UAVs and MS in the post-disaster stage should attract much more concern at present.

For such motivations and research gaps, this paper proposed a multi-period distributionally robust resilience enhancement model (MPDRRM) for a TDCS, and utilized UAVs to address communication

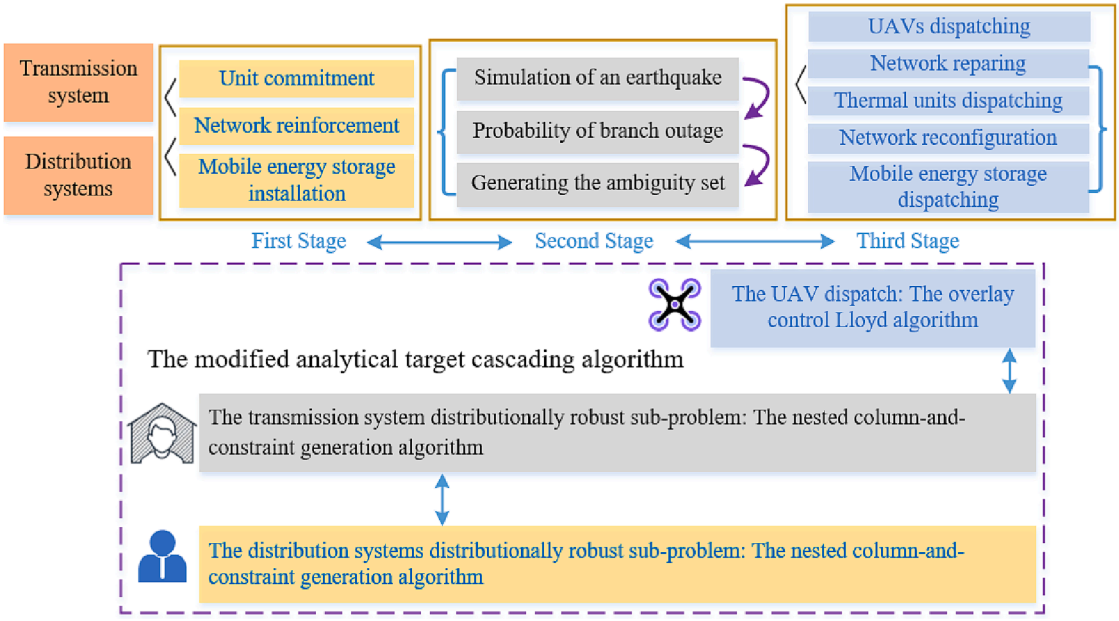


Fig. 2. The architecture of the proposed model and algorithms.

failures under natural disaster attacks and support the information interaction. The proposed model can be generally regarded as a common defender-attacker-defender (DAD) model, which entirely considers the active reinforcement of the TDCS before the disaster, the most serious attacks during the disaster, and the restoration of the TDCS after the disaster. To comprehensively consider the uncertainties of a natural disaster and efficiently solve the proposed MPDRRM, the natural disaster model with earthquakes as an example is proposed, and the distributionally robust model with an ambiguity set that incorporates both moment and Wasserstein metric information of uncertain contingencies [40] is applied in this paper. The ATC algorithm embedded with a nested column-and-constraint generation (CCG) algorithm [41] is applied to solve the proposed model. The framework of this model is proposed in Fig. 1, and the relationship between several sub-systems is also declared in it. According to Fig. 1, three contributions are clarified below follows.

(i) We propose a resilient-oriented collaborative planning and operation method for the TDCS, which can offer flexible planning and operating solutions for all transmission and distribution systems to support each other when natural disasters occur.
(ii) To recover the communication system between the TS and several DSs, the model of UAVs' communication coverage area is applied. Then, a resilience-oriented UAV optimal dispatch model is proposed as a multi-period coverage function optimization problem in this paper, and an overlay control Lloyd (OCL) algorithm is applied to solve this model at each time period.
(iii) To address the uncertainties of natural disasters, an MPDRRM is formulated for the TDCS, which takes the network reinforcement, the most severe attack, the network restoration, and UAV dispatch into consideration. The ambiguity set of the DR model is constructed based on the earthquake damage model and incorporates both the moment and the Wasserstein metric formation of uncertain contingencies. The MPDRRM can be decomposed by several sub-problems, and solved by an ATC algorithm embedded with a nested CCG algorithm. To give a whole comprehension of the proposed model and contributions, the architecture of the proposed model and solving algorithms for TDCSs considering UAVs are declared in Fig. 2.

The organization of this paper can be summarized as follows. Section 2 details the modeling of UAVs and the related multi-period dispatch scheme for the information part of the TDCS. Section 3 details the modeling of earthquakes and the physical part of the TDCS. Section 4 presents the solution methodology. Section 5 provides the case studies for the proposed model. Section 6 concludes this paper.

Before providing the model formulation, we have to outline some key considerations and assumptions as follows.

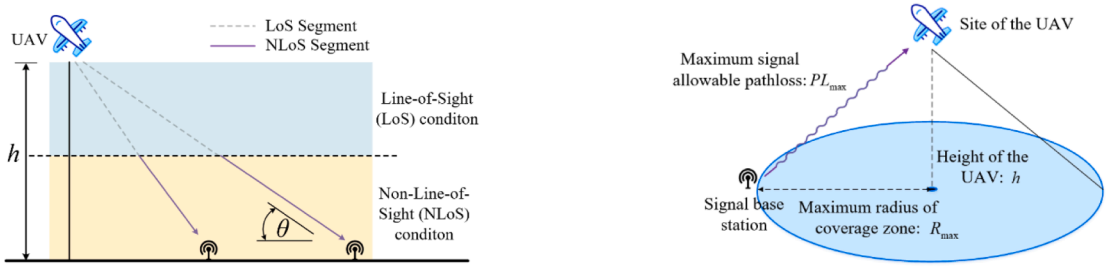
(1) We concentrate on earthquakes because not only it is the primary and one of the most common natural disasters, but also UAVs can be effectively implemented to support communication systems.
(2) We assume that the movement time and charging time for a UAV can also be ignored. This assumption is based on the current technological development of UAVs, and the movement time of a UAV in a TS or DS will be much less than the dispatch period.
(3) We assume that only when the earthquake completely stops can the network restoration be executed. The set of dispatching periods can be expressed as $\Omega^{D} = \{t \mid 1 \le t \le T^{SS}\} \cup \{t \mid T^{SS} \le t \le T^{TS}\}$.
(4) We assume that if the barycenter of a branch falls in the coverage zone of the UAVs, the communication system of this branch can be restored, which means that the restoration of this branch can be dispatched at this period.
(5) For any matrix A, the transpose of A is represented as A^{T}.

2. Modeling of the UAVs

In this section, a model of UAV is first proposed. Based on the maximum coverage zone for a UAV, an optimal dispatch model of UAVs for one period is formulated, and the overlay control Lloyd algorithm is applied to access the optimal coverage position of UAVs. Finally, we extend the single-time dispatch model of UAV into multiple periods, and the information part of the MPDRRM is proposed.

2.1. Maximum coverage zone for a UAV

In recent years, wireless networks are increasingly adopted by users for critical mission communications (i.e., public safety agencies). And since current communication is largely dependent on fixed



(a) Two segments in the wireless signal transmission

(b) The coverage zone of a UAV

Fig. 3. The physical model of a UAV.

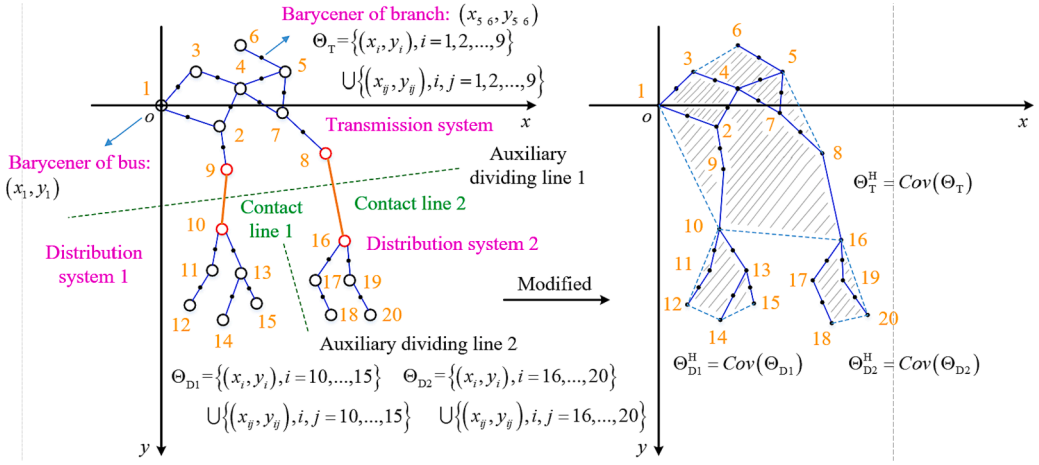


Fig. 4. The coordinate system for a simple TDCS and the modified process.

infrastructure (such as base stations and towers), the communication systems can be severally disrupted by natural disasters, and the airborne communication infrastructure concept has been gradually endorsed in many countries. Speaking of which, UAVs gradually attract much more concern for their flexibility and efficiency. As a typical type of low-altitude platform, UAVs are much easier to deploy with the broadband cellular concept. The technology carried out by UAVs can depend on the end user's application, budget, and bandwidth requirements. Applications could be as advanced as LTE-A, Wi-Fi, and WiMAX, or as legacy as GSM, TETRA, or P-25 systems.

Acting as an airborne base station, the key to dispatching UAVs concentrates on the deployment position, so that more users and faulted lines can be reached by the temporary communication network. However, as the wireless transmission of signals will attenuate in the transmission medium, there will always exist a maximum value for the effective signal coverage of UAVs in a specific urban environment. In recent years, many papers have been proposed about wireless signal transmission attenuation problems. The most comprehensive work about the wireless signal transmission model can be found in [42], and it divides the medium in which the signal is transmitted wirelessly into two segments with different attenuation efficiencies, which is shown in Fig. 3(a). The coverage area of a UAV can be regarded as a cone with a variable radius at the bottom, which is shown in Fig. 3(b). When the maximum signal attenuation is determined, the maximum coverage can be determined by adjusting the height of the UAV.

As shown in Fig. 3(a), the transmission medium of the wireless signal transmission between a user and a UAV can be divided into two segments: the Line-of-Sight (LoS) segment and the non-Line-of-Sight (NLoS) segment. Generally, the signals emitted by a UAV will first propagate in free space where the power attenuation is relatively low until reaching the urban environment where the signals will incur shadowing and scattering caused by man-made structures. The transmission medium in the free space is called the LoS medium, and the transmission medium in the urban environment is called the NLoS medium. The signal loss in the LoS condition is defined as the free space pathloss (FSPL), and the signal loss in the NLoS condition is the excessive pathloss. In general, the pathloss of a wireless signal emitted from a UAV to a user contains both the FSPL and the excessive pathloss.

Fig. 3 illustrates that the signal pathloss is mainly related to the height and the coverage zone of a UAV in a specific signal transmission environment. To ensure the non-distortion of the signal, there usually exists an upper bound of the pathloss, which is also called the maximum signal allowable path loss (MSAP). Once the MSAP is determined, the coverage zone of a UAV can be determined by the height of this UAV. According to [42], the coverage zone of a UAV can be calculated through the parameter of the maximum signal allowable path loss (MSAP) $PL_{\max}$ and the altitude of this UAV. The relationship between the MSAP, altitude, and radius of the coverage zone for a UAV can be expressed as

$$\begin{cases} PL_{\max} = \dfrac{A}{1 + a\exp(-b[\arctan(h/R) - a])} + 10\log(h^2 + R^2) + B \\ A = \eta_{\text{LoS}} - \eta_{\text{NLoS}} \\ B = 20\log f + 20\log(4\pi/c) + \eta_{\text{NLoS}} \end{cases} \tag{1}$$

where a and b are S-curve parameters [42], which are constants in this function. h and R denote the altitude and the radius of the coverage zone for a UAV. f denotes the communication frequency constant, and c is the light speed. η_{LoS} and η_{NLoS} denote the excessive path loss in a Line-of-Sight condition and a near-Line-of-Sight condition. These two parameters are two constants related to urban environments.

Generally, the MASP can be approximately regarded as a constant under certain urban environments [42]. Thus, the radius of the coverage

zone for a UAV is only related to its altitude, and the maximum radius of the coverage zone $R_{\max}$ can also be calculated through function (1).

2.2. Optimal dispatch of UAVs for a single period

After a natural disaster, the dispatch of UAVs can be treated as a multi-period coverage model. In each period, the dispatch of UAVs has the same target. We first take a single period as an example to analyze the UAV dispatching model in this part.

The optimal dispatching of UAVs is regarded as a coverage function optimization problem. We first establish a Planar Cartesian Coordinate System covering the top view of the TDCS as shown in Fig. 4. To be convenient, we define two groups of mappings, representing the correspondence from the set of bus numbers to the set of bus coordinates and the correspondence from the set of branch numbers to the set of branch coordinates:

$$f^{\mathrm{BS}}:\Xi^{\mathrm{BS}}\to\Gamma^{\mathrm{BS}} \tag{2a}$$

$$f^{\mathrm{CBR}}:\Xi^{\mathrm{CBR}}\to\Gamma^{\mathrm{CBR}} \tag{2b}$$

where Ξ^{BR} and Ξ^{CBR} are the sets of bus numbers and branch numbers. Γ^{BS} and Γ^{CBR} denote the sets of bus barycenter coordinates and branch barycenter coordinates. Thus, each bus and transmission line can be given a certain coordinate.

Consider that there are χ different UAVs, containing χ^t and χ^d UAVs respectively deployed in the TS and DSs, utilized to form the temporary communication system and perform the communication coverage tasks. The dispatch of UAVs at each period is regarded as a coverage function optimization problem [43], which can be expressed as

$$\begin{aligned}
&\min_{\sigma,W} H(\mathrm{P},W)=H_{\mathrm{T}}(\mathrm{P},W)+\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}H_{\mathrm{D}_l}(\mathrm{P},W)\\
&H_{\mathrm{T}}(\mathrm{P},W)=\sum_{m\in\mathscr{N}_{\mathrm{UT}}}\sum_{z\in W_m}g(z,\sigma_m)\varphi(z)\\
&H_{\mathrm{D}_l}(\mathrm{P},W)=\sum_{k\in\mathscr{N}_{\mathrm{UD_l}}}\sum_{z\in W_k}g(z,\sigma_k)\varphi(z),\forall l\in\mathscr{N}_{\mathrm{n}}\\
&s.t.\ z\in\Theta=\Theta_{\mathrm{T}}\cup\Theta_{\mathrm{D}},\ \Theta_{\mathrm{D}}=\bigcup\nolimits_{\forall l\in\mathscr{N}_{\mathrm{n}}}\Theta_{\mathrm{D}_l}\\
&\sigma_m\in\Theta_{\mathrm{T}},\ \forall m\in\mathscr{N}_{\mathrm{UT}}\\
&\sigma_k\in\Theta_{\mathrm{D}},\ \forall k\in\bigcup\nolimits_{\forall l\in\mathscr{N}_{\mathrm{n}}}\mathscr{N}_{\mathrm{UD_l}}\\
&W_j=\left\{z\in\Theta_{\mathrm{T}}\cup\Theta_{\mathrm{D}}\middle|\|z-\sigma_j\|_2^2\leqslant R_j^{\max}\right\},\forall j\in\mathscr{N}_{\mathrm{U}}\\
&W=\left\{W_j\middle|\forall j\in\mathscr{N}_{\mathrm{U}}\right\}
\end{aligned} \tag{3a}$$

where z denotes the barycenter coordinate of each element in the TDCS. $\mathrm{P}=\{\sigma_j=(\sigma_j^x,\sigma_j^y),j=1,2,\ldots,\chi\}$ denotes the set of the barycenter coordinate of UAVs in the TDCS. z denotes the barycenter coordinate of each element in the TDCS. $W=\{W_j|\ \forall j\in\mathscr{N}_{\mathrm{U}}\}$is the set of coverage zone of UAV. $\Theta=\Theta_{\mathrm{T}}\cup\Theta_{\mathrm{D}}$ denotes the set of barycenter coordinates of all buses and lines in the TDCS. Θ_{T} and $\Theta\ \mathrm{D}_l(l=1,2,\ldots,n)$ denote the sets of barycenter coordinates of buses and lines in the TS and the i-th DS. $\mathscr{N}_{\mathrm{n}}=\{1,2,\ldots,n\}$ is the set of distribution systems. $\mathscr{N}_{\mathrm{U}}=\{1,2,\ldots,\chi\}$is the set of UAVs. $\mathscr{N}_{\mathrm{UT}}$ is the set of UAVs applied in the transmission system. $\mathscr{N}_{\mathrm{UD}_l}$ is the set of UAVs applied in the l-th distribution system. $\varphi(z)$ is so-called the environmental density function (EDF) which can determine the perceptual performance of UAVs in performing optimal coverage. Based on the optimization model proposed in (2a), we can adopt optimal deployment of UAVs for different purposes by varying the EDFs, and this function will be discussed later.

Mathematically, function g: $R_+\to R_+$ in (3a) should be a non-decreasing function and can be defined as the Euclidean distance between two points, which can be expressed below.

$$g(z,\sigma_j)=\begin{cases}\|z-\sigma_j\|_2^2,\forall z\in W_j,j\in\mathscr{N}_{\mathrm{U}}\\ 0,\forall z\notin W_j,j\in\mathscr{N}_{\mathrm{U}}\end{cases} \tag{3b}$$

Note that (3) is a non-convex optimization problem. We also cannot guarantee that Θ will be a convex set for every TDCS. Thus, the optimization problem in (3) cannot be solved directly, leading to the application of the overlay control Lloyd (OCL) algorithm in this paper.

To guarantee the convergence of the OCL algorithm, we should ensure that the definition field of z is a convex set [43]. Hence, convex hulls of sets Θ_{T} and Θ_{D}, which are the smallest convex sets that respectively contain Θ_{T} and Θ_{D}, are applied to modify (3).

$$\begin{aligned}
&\min_{\sigma}\widetilde{H}\left(\mathrm{P},\widetilde{W}\right)=\widetilde{H}_{\mathrm{T}}\left(\mathrm{P},\widetilde{W}\right)+\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\widetilde{H}_{\mathrm{D}_l}\left(\mathrm{P},\widetilde{W}\right)\\
&\widetilde{H}_{\mathrm{T}}\left(\mathrm{P},\widetilde{W}\right)=\sum_{m\in\mathscr{N}_{\mathrm{UT}}}\sum_{z\in\widetilde{W}_m}\widetilde{g}(z,\sigma_m)\widetilde{\varphi}(z)\\
&\widetilde{H}_{\mathrm{D}_l}\left(\mathrm{P},\widetilde{W}\right)=\sum_{k\in\mathscr{N}_{\mathrm{UD_l}}}\sum_{z\in\widetilde{W}_k}\widetilde{g}(z,\sigma_k)\widetilde{\varphi}(z),\forall l\in\mathscr{N}_{\mathrm{n}}\\
&\widetilde{g}(z,\sigma_j)=\begin{cases}\|z-\sigma_j\|_2^2,\forall z\in\widetilde{W}_j,j\in\mathscr{N}_{\mathrm{U}}\\ 0,\forall z\notin\widetilde{W}_j,j\in\mathscr{N}_{\mathrm{U}}\end{cases}\\
&s.t.\ z\in\Theta^{\mathrm{H}}=\Theta_{\mathrm{T}}^{\mathrm{H}}\cup\Theta_{\mathrm{D}}^{\mathrm{H}},\\
&\Theta_{\mathrm{T}}^{\mathrm{H}}=Conv(\Theta_{\mathrm{T}}),\ \Theta_{\mathrm{D}}^{\mathrm{H}}=\bigcup\nolimits_{\forall l\in\mathscr{N}_{\mathrm{n}}}\Theta_{\mathrm{D}_l}^{\mathrm{H}}=\bigcup\nolimits_{\forall l\in\mathscr{N}_{\mathrm{n}}}Conv(\Theta_{\mathrm{D}_l})\\
&\sigma_m\in\Theta_{\mathrm{T}}^{\mathrm{H}},\ \forall m\in\mathscr{N}_{\mathrm{UT}}\\
&\sigma_k\in\Theta_{\mathrm{D}}^{\mathrm{H}},\ \forall k\in\bigcup\nolimits_{\forall l\in\mathscr{N}_{\mathrm{n}}}\mathscr{N}_{\mathrm{UD_l}}\\
&\widetilde{W}_j=\left\{z\in\Theta_{\mathrm{T}}^{\mathrm{H}}\cup\Theta_{\mathrm{D}}^{\mathrm{H}}\middle|\|z-\sigma_j\|_2^2\leqslant R_j^{\max}\right\},\forall j\in\mathscr{N}_{\mathrm{U}}\\
&\widetilde{W}=\left\{\widetilde{W}_j\middle|\forall j\in\mathscr{N}_{\mathrm{U}}\right\}
\end{aligned} \tag{4a}$$

where $\Theta_{\mathrm{T}}^{\mathrm{H}}$ and $\Theta_{\mathrm{D}}^{\mathrm{H}}$ are the convex hulls of sets Θ_{T} and Θ_{D}, and $\widetilde{\varphi}$ is the modified EDF.

Then, the optimal partition of $\Theta^{\mathrm{H}}=\Theta_{\mathrm{T}}^{\mathrm{H}}\cup\Theta_{\mathrm{D}}^{\mathrm{H}}$ can be divided into several Voronoi cells (i.e., Vor_1, Vor_2, …, Vor_χ) by applying the Voronoi partition, which can be expressed as

$$\begin{aligned}
&Vor(\mathrm{P})=\left\{Vor_j\middle|\forall j\in\mathscr{N}_{\mathrm{U}}\right\}\\
&Vor_j=\left\{z\in\Theta^{\mathrm{H}}\middle|\|z-\sigma_i\|_2^2\geqslant\|z-\sigma_j\|_2^2,\forall i\in\mathscr{N}_{\mathrm{U}},\mathrm{i}\neq\mathrm{j}\right\},\forall j\in\mathscr{N}_{\mathrm{U}}
\end{aligned} \tag{4b}$$

This division makes the points assigned to the j-th UAV in Θ^{H} the closest to it. More details about the generation of Voronoi cells can be referred to [44–47]. By applying the Voronoi partition, the coverage function optimization problem shown in (4a) can be further modified into

$$\begin{aligned}
&\min_{\sigma}\widetilde{H}_{\mathrm{V}}(\mathrm{P})=\widetilde{H}_{\mathrm{V}}^{\mathrm{T}}(\mathrm{P})+\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\widetilde{H}_{\mathrm{V}}^{\mathrm{D}_l}(\mathrm{P})\\
&\widetilde{H}_{\mathrm{V}}^{\mathrm{T}}(\mathrm{P})=\sum_{m\in\mathscr{N}_{\mathrm{UT}}}\sum_{z\in Vor_m}\widetilde{g}(z,\sigma_m)\widetilde{\varphi}(z)\\
&\widetilde{H}_{\mathrm{V}}^{\mathrm{D}_l}(\mathrm{P})=\sum_{k\in\mathscr{N}_{\mathrm{UD_l}}}\sum_{z\in Vor_k}\widetilde{g}(z,\sigma_k)\widetilde{\varphi}(z),\forall l\in\mathscr{N}_{\mathrm{n}}\\
&s.t.\ z\in\Theta^{\mathrm{H}},\ \sigma_j\in\Theta^{\mathrm{H}},\ \forall j\in\mathscr{N}_{\mathrm{U}}\\
&Vor_j=\left\{z\in\Theta^{\mathrm{H}}\middle|\|z-\sigma_i\|_2^2\geqslant\|z-\sigma_j\|_2^2,\forall i\in\mathscr{N}_{\mathrm{U}},\mathrm{i}\neq\mathrm{j}\right\},\forall j\in\mathscr{N}_{\mathrm{U}}
\end{aligned} \tag{4c}$$

We only need to solve $\widetilde{H}_{\mathrm{V}}(\mathrm{P})$ in (4c) to get the optimal position of each UAV. By differentiating $\widetilde{H}_{\mathrm{V}}(\mathrm{P})$, we can get the following functions

$$\begin{cases} \partial\widetilde{H}_{\mathrm{V}}(\mathrm{P})\Big/\partial\sigma_j = 2MV_j(\sigma_j - CV_j), \forall j \in \mathscr{N}_{\mathrm{U}} \\ CV_j = LV_j/MV_j, \forall j \in \mathscr{N}_{\mathrm{U}} \\ MV_j = \sum_{\forall z \in Vor_j} \widetilde{\varphi}(z), \forall j \in \mathscr{N}_{\mathrm{U}} \\ LV_j = \sum_{\forall z \in Vor_j} z\widetilde{\varphi}(z), \forall j \in \mathscr{N}_{\mathrm{U}} \end{cases} \tag{4d}$$

where MV_j, LV_j, and CV_j denote the generalized mass, polar moment of inertia, and the barycenter of the j-th Voronoi cell. When σ_j is the barycenter of Vor_j, the (4c) has the minimum value, which can be expressed as

$$CV_j = \underset{\sigma_j}{\operatorname{argmin}} \widetilde{H}_{\mathrm{V}}(\mathrm{P}), \forall j \in \mathscr{N}_{\mathrm{U}} \tag{5}$$

This means that we only need to calculate the barycenter coordinates of Vor_j to find the optimal location of each UAV. Then, the OCL algorithm is applied to make each UAV realize the barycenter-Voronoi configuration, which can get the optimal solution for (4c).

Assume that the original position of the j-th UAV is $\sigma_j(0)$. The OCL algorithm can be executed by the following flowchart [43] and get the optimal dispatch of UAV at each period.

2.3. Multi-period dispatch model for UAVs

The modified coverage function optimization problem in each period with the Voronoi partition can be expressed as follows:

$$\begin{aligned} \min_{\sigma} \widetilde{H}_{\mathrm{V}}(\mathrm{P},t) &= \widetilde{H}_{\mathrm{V}}^{\mathrm{T}}(\mathrm{P},t) + \sum_{\forall l \in \mathscr{N}_{\mathrm{n}}} \widetilde{H}_{\mathrm{V}}^{\mathrm{D}_l}(\mathrm{P},t) \\ \widetilde{H}_{\mathrm{V}}^{\mathrm{T}}(\mathrm{P},t) &= \sum_{m \in \mathscr{N}_{\mathrm{UT}}} \sum_{z \in Vor_m} \widetilde{g}\left(z,\sigma_{m,t}\right)\widetilde{\varphi}_t(z), \forall t \geqslant T^{\mathrm{SS}} \\ \widetilde{H}_{\mathrm{V}}^{\mathrm{D}_l}(\mathrm{P},t) &= \sum_{k \in \mathscr{N}_{\mathrm{UD_l}}} \sum_{z \in Vor_k} \widetilde{g}(z,\sigma_k)\widetilde{\varphi}_t(z), \forall t \geqslant T^{\mathrm{SS}} \\ s.t.\ z \in \Theta^{\mathrm{H}},\ \sigma_{j,t} \in \Theta^{\mathrm{H}},\ \forall j \in \mathscr{N}_{\mathrm{U}},\ \mathrm{t} &\geqslant \mathrm{T}^{\mathrm{SS}} \\ Vor_{j,t} = \left\{ z \in \Theta^{\mathrm{H}} \middle| \left\|z-\sigma_{i,t}\right\|_2^2 \geqslant \left\|z-\sigma_{j,t}\right\|_2^2, \forall i \in \mathscr{N}_{\mathrm{U}}, \mathrm{i} \neq \mathrm{j} \right\}, \forall j \in \mathscr{N}_{\mathrm{U}},\ \mathrm{t} &\geqslant \mathrm{T}^{\mathrm{SS}} \\ \widetilde{W}_{j,t} = \left\{ z \in \Theta^{\mathrm{H}} \middle| \left\|z-\sigma_{j,t}\right\|_2^2 \leqslant R_j^{\max} \right\}, \forall j \in \mathscr{N}_{\mathrm{U}},\ \forall \mathrm{t} &\geqslant \mathrm{T}^{\mathrm{SS}} \end{aligned} \tag{6a}$$

By changing the EDF $\varphi(z)$, we can get different optimal dispatch results of UAVs for different purposes. In this paper, a certain purpose of the proposed model is to improve power recovery after a natural disaster. To this end, the UAVs should cover as many faulted lines and buses as possible. Thus, based on the power flow in the normal operation, we can define an EDF in the following way

$$\widetilde{\varphi}_0(z) = \sum_{d \in \Theta^{\mathrm{H}}} \mathscr{A}_{d,0} e^{-M\|z-z_d\|}, \forall z \in \Theta^{\mathrm{H}}$$

$$\mathscr{A}_{z,0} = \begin{cases} -\lambda_z P_z^{\mathrm{D}} \sum_{\forall x \in \Psi(z)} p_x^{\mathrm{CBR}}, \forall z \in \Gamma^{\mathrm{BS}} \\ -p_z^{\mathrm{CBR}} P_z^{\mathrm{T}}, \forall z \in \Gamma^{\mathrm{CBR}} \\ 0, \forall z \in \Theta^{\mathrm{H}}, z \notin \Gamma^{\mathrm{BS}} \cup \Gamma^{\mathrm{CBR}} \end{cases} \tag{6b}$$

where λ_z is the priority of the power load at bus z. p_z^{CBR} is the branch outage probability during a natural disaster and will be discussed later. PD_z and PT_z are the power load at bus z and power flow in branch z at normal state. $\Psi(z)$ is the set of branches that are connected to bus z. M is a constant with a large value.

The faulted lines can be gradually repaired, which means the EDF keeps changing at different periods. The EDF can be expressed as

$$\widetilde{\varphi}_t(z) = \sum_{d \in \Theta^{\mathrm{H}}} \mathscr{A}_{d,t} e^{-M\|z-z_d\|}, \forall z \in \Theta^{\mathrm{H}}, t \geqslant T^{\mathrm{SS}}$$

$$\mathscr{A}_{z,t} = \begin{cases} -\lambda_z P_z^{\mathrm{D}} \sum_{\forall x \in \Psi(z)} \widetilde{p}_x^{\mathrm{CBR}}(t), \forall z \in \Gamma^{\mathrm{BS}} \\ -\widetilde{p}_z^{\mathrm{CBR}}(t) P_z^{\mathrm{T}}, \forall z \in \Gamma^{\mathrm{CBR}} \\ 0, \forall z \in \Theta^{\mathrm{H}}, z \notin \Gamma^{\mathrm{BS}} \cup \Gamma^{\mathrm{CBR}} \end{cases} \tag{6c}$$

where $\widetilde{p}_z^{\mathrm{CBR}}(t)$ is an auxiliary function about transmission line z and can be expressed as

$$\widetilde{p}_z^{\mathrm{CBR}}(t) = p_{z,T^{SS}}^{\mathrm{CBR}} - x_{z,t}, \forall z \in \Gamma^{\mathrm{CBR}}, t \geqslant T^{\mathrm{SS}} \tag{6d}$$

where $x_{z,t}$ is a binary variable indicating the branch outage state: $x_{z,t} = 0$ if outage; 1, otherwise.

By solving (6a)-(6d) and applying the OCL algorithm from Step O-1 to Step O-9, we can get the optimal UAV dispatch results in each dispatching period.

Without changing the optimization results and the process of the OCL algorithm, an ATC-pertained penalty function is now added to (6a), which can be expressed as (6e).

$$\begin{aligned} \min_{\sigma} \widetilde{H}_{\mathrm{V}}(\mathrm{P},t) &= \widetilde{H}_{\mathrm{V}}^{\mathrm{T}}(\mathrm{P},t) + \sum_{\forall l \in \mathscr{N}_{\mathrm{n}}} \widetilde{H}_{\mathrm{V}}^{\mathrm{D}_l}(\mathrm{P},t) \\ &+ \left[\begin{array}{l} \sum_{\forall l \in \mathscr{N}_{\mathrm{n}}} \left[\kappa_{l,t}^{\mathrm{U}} \left(\mathrm{Z}_t^{\mathrm{ID}_l} - \widetilde{\mathrm{Z}}_t^{\mathrm{PD}_l} \right) + \zeta_{l,t}^{\mathrm{U}} \left(\mathrm{Z}_t^{\mathrm{ID}_l} - \widetilde{\mathrm{Z}}_t^{\mathrm{PD}_l} \right)^2 \right] \\ + \left[\kappa_{n+1,t}^{\mathrm{U}} \left(\mathrm{Z}_t^{\mathrm{IT}} - \widetilde{\mathrm{Z}}_t^{\mathrm{PT}} \right) + \zeta_{n+1,t}^{\mathrm{U}} \left(\mathrm{Z}_t^{\mathrm{IT}} - \widetilde{\mathrm{Z}}_t^{\mathrm{PT}} \right)^2 \right] \end{array} \right], \forall t \geqslant T^{\mathrm{SS}} \\ \mathrm{Z}_t^{\mathrm{IT}} = \sum_{j \in \mathscr{N}_{\mathrm{UT}}} \sum_{z \in \widetilde{W}_{j,t}} \varphi_t^{\mathrm{I}}(z),\ \widetilde{\mathrm{Z}}_t^{\mathrm{PT}} &= \sum_{j \in \mathscr{N}_{\mathrm{UT}}} \sum_{z \in \widetilde{W}_{j,t}} \widetilde{\varphi}_t^{\mathrm{P}}(z), \\ \mathrm{Z}_t^{\mathrm{ID}_l} = \sum_{j \in \mathscr{N}_{\mathrm{UD_l}}} \sum_{z \in \widetilde{W}_{j,t}} \varphi_t^{\mathrm{I}}(z),\ \widetilde{\mathrm{Z}}_t^{\mathrm{PD}_l} &= \sum_{j \in \mathscr{N}_{\mathrm{UD_l}}} \sum_{z \in \widetilde{W}_{j,t}} \widetilde{\varphi}_t^{\mathrm{P}}(z), \quad \forall l \in \mathscr{N}_{\mathrm{n}},\ \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \\ s.t.\ z \in \Theta^{\mathrm{H}},\ \sigma_{j,t} &\in \Theta^{\mathrm{H}},\ \forall j \in \mathscr{N}_{\mathrm{U}},\ \forall \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \\ Vor_{j,t} = \left\{ z \in \Theta^{\mathrm{H}} \middle| \left\|z-\sigma_{i,t}\right\|_2^2 \geqslant \left\|z-\sigma_{j,t}\right\|_2^2, \forall i \in \mathscr{N}_{\mathrm{U}}, \mathrm{i} \neq \mathrm{j} \right\}, &\forall j \in \mathscr{N}_{\mathrm{U}},\ \forall \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \\ \widetilde{W}_{j,t} = \left\{ z \in \Theta^{\mathrm{H}} \middle| \left\|z-\sigma_{j,t}\right\|_2^2 \leqslant R_j^{\max} \right\}, &\forall j \in \mathscr{N}_{\mathrm{U}},\ \forall \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \end{aligned} \tag{6e}$$

where $\kappa_{l,t}^{\mathrm{U}}$ and $\zeta_{l,t}^{\mathrm{U}}$ are the penalty function multipliers utilized in the ATC algorithm and will be discussed later. (6e) can also be regarded as the information part of the proposed MPDRRM.

Focus on several EDFs φ_t^{I}, φ_t^{P}, $\widetilde{\varphi}_t^{\mathrm{I}}$, $\widetilde{\varphi}_t^{\mathrm{P}}$ applied in the penalty functions in (6e). These EDFs are applied in different sub-problems of the TDCS while implementing the ATC algorithm. φ_t^{I}, φ_t^{P} in (6d) are applied in the information sub-problem and physical sub-problem, relatively. $\widetilde{\varphi}_t^{\mathrm{I}}$, $\widetilde{\varphi}_t^{\mathrm{P}}$ are the calculation results by applying the optimization results of the information sub-problem and physical sub-problem to (6e) in the iterative process, the details of which will be discussed in Section IV.

3. Modeling of the physical part of the TDCS

In this section, the model of earthquakes and the branch outage probability are constructed to derive the ambiguity set for the DR model. Then, the physical part of the MPDRRM is set up to give a resilient-oriented collaborative planning and operation approach for the TDCS.

3.1. First stage: Network reinforcement

Hardening transmission lines and deploying MS are emergent investment approaches to resist earthquakes. The decisions and operation

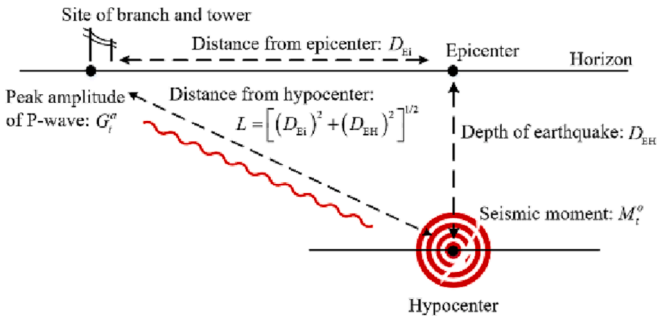


Fig. 5. Modeling of an earthquake.

constraints about the optimal day-ahead unit commitment in the TS are also included in this stage. The objective function is shown in (7a), containing the cost of hardening transmission lines, deploying MS, and unit commitment. The power and energy capacities of MS are bounded in (7b)–(7c). The line hardening decisions are constrained by (7d)–(7e). The constraints of unit start-up/shut-down are shown in (7f)–(7h).

$$\psi_{\mathrm{FS}}(s)=\phi_{\mathrm{GEN}}+\min_{s_{\mathrm{p}}^{\mathrm{FS}}}\sup_{Y\in F}E_{\mathrm{Y}}(\phi_{\mathrm{FS}})$$

$$\phi_{\mathrm{GEN}}=\min_{s_{\mathrm{u}}^{\mathrm{FS}}}\sum_{t\geqslant T^{\mathrm{SS}}}\sum_{g\in\Xi^{\mathrm{ther}}}\left(c_g^{\mathrm{SU}}u_{g,t}+c_g^{\mathrm{SD}}v_{g,t}+c_g^{\mathrm{NL}}w_{g,t}\right)$$

$$\phi_{\mathrm{FS}}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)=\phi_{\mathrm{FS}}^{\mathrm{T}}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)+\phi_{\mathrm{FS}}^{\mathrm{D}}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)$$

$$\phi_{\mathrm{FS}}^{\mathrm{T}}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)=\min_{s^{\mathrm{TS}}}\left(\sum_{(i,j)\in\Xi_{\mathrm{har}}^{\mathrm{TCBR}}}c_{ij}^{\mathrm{harT}}x_{ij}^{\mathrm{harT}}\right)$$

$$\phi_{\mathrm{FS}}^{\mathrm{D}}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)=\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\phi_{\mathrm{FS}}^{\mathrm{D}_l}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)$$

$$\phi_{\mathrm{FS}}^{\mathrm{D}_l}\left(s_{\mathrm{p}}^{\mathrm{FS}},s^{\mathrm{TS}}\right)=\min_{s^{\mathrm{TS}}}\left(\begin{array}{c}\sum\limits_{(i,j)\in\Xi_{\mathrm{har}}^{\mathrm{DCBR}_l}}c_{ij}^{\mathrm{harD}_l}x_{ij}^{\mathrm{harD}_l}\\+\sum\limits_{i\in\delta_l^{\mathrm{MS}}}\left(c_{i,l}^{\mathrm{pcap}}P_{i,l}^{\mathrm{SMax}}+c_{i,l}^{\mathrm{ecap}}E_{i,l}^{\mathrm{SMax}}\right)\end{array}\right),\forall l\in\mathscr{N}_{\mathrm{n}} \tag{7a}$$

$$s.t.\ 0\leqslant P_{i,l}^{\mathrm{SMax}}\leqslant\overline{P}_{i,l}^{\mathrm{SMax}},\ \forall i\in\delta_l^{\mathrm{MS}},l\in\mathscr{N}_{\mathrm{n}} \tag{7b}$$

$$0\leqslant E_{i,l}^{\mathrm{SMax}}\leqslant\overline{E}_{i,l}^{\mathrm{SMax}},\ \forall i\in\delta_l^{\mathrm{MS}},l\in\mathscr{N}_{\mathrm{n}} \tag{7c}$$

$$\begin{cases}x_{ij}^{\mathrm{harT}}\leqslant x_{ij,t},\forall t\leqslant T^{\mathrm{TS}},(i,j)\in\Xi_{\mathrm{T}}^{\mathrm{CBR}}\\\sum\limits_{(i,j)\in\Xi_{\mathrm{har}}^{\mathrm{TCBR}}}x_{ij}^{\mathrm{harT}}\leqslant n_{\max}^{\mathrm{T}},\forall(i,j)\in\Xi_{\mathrm{T}}^{\mathrm{CBR}}\end{cases} \tag{7d}$$

$$\begin{cases}x_{ij}^{\mathrm{harD}_l}\leqslant x_{ij,t},\forall t\leqslant T^{\mathrm{TS}},(i,j)\in\Xi_{\mathrm{D}_l}^{\mathrm{CBR}},\forall l\in\mathscr{N}_{\mathrm{n}}\\\sum\limits_{(i,j)\in\Xi_{\mathrm{har}}^{\mathrm{CBRD}_l}}x_{ij}^{\mathrm{harD}_l}\leqslant n_{\max}^{\mathrm{D}_l},\forall l\in\mathscr{N}_{\mathrm{n}}\end{cases} \tag{7e}$$

$$w_{g,t}-w_{g,t-1}=u_{g,t}-v_{g,t},\ \forall t\geqslant T^{\mathrm{SS}},g\in\Xi^{\mathrm{ther}} \tag{7f}$$

$$w_{g,t}\geqslant u_{g,t},\forall\tau\leqslant t\leqslant\tau+UT_g-1,\tau\geqslant T^{\mathrm{SS}},g\in\Xi^{\mathrm{ther}} \tag{7g}$$

$$1-w_{g,t}\geqslant v_{g,t},\forall\tau\leqslant t\leqslant\tau+DT_g-1,\tau\geqslant T^{\mathrm{SS}},g\in\Xi^{\mathrm{ther}} \tag{7h}$$

where $s_{\mathrm{u}}^{\mathrm{FS}}=\{u_{g,t},v_{g,t},w_{g,t}\}$ denotes the unit commitment variables for the thermal unit (TU) g in the TS determined in the first stage, and $s_{\mathrm{p}}^{\mathrm{FS}}=\{x_{ij}^{\mathrm{har}},P_{i,l}^{\mathrm{SMax}},E_{i,l}^{\mathrm{SMax}}\}$ denotes the planning variables determined in the first stage. x_{ij}^{harT} and $x_{ij}^{\mathrm{harD}}=\{x_{ij}^{\mathrm{harD}_l},l=1,\ldots,n\}$ are the binary variables indicating whether hardening the candidate branch (i,j) in the TS and DSs. $u_{g,t}/v_{g,t}/w_{g,t}$ denote whether TU g is start-up/shut-down/on at period t, and $P_{i,l}^{\mathrm{SMax}}/E_{i,l}^{\mathrm{SMax}}$ denote the maximum power capacity and energy capacity for the MS i. δ_l^{MS} is the set of MS invested in the l-th distribution system. Constants $c_{ij}^{\mathrm{harT}}/c_{ij}^{\mathrm{harD}_l}/c_{i,l}^{\mathrm{pcap}}/c_{i,l}^{\mathrm{ecap}}/c_g^{\mathrm{SU}}/c_g^{\mathrm{SD}}/c_g^{\mathrm{NL}}$ are the line hardening cost, MS investment cost, and TU start-up/shut-down/no-load cost, respectively. $\Xi_{\mathrm{har}}^{\mathrm{TCBR}}$ and $\Xi_{\mathrm{har}}^{\mathrm{DCBR}_l}$ are the sets of candidate hardened branches in the TS and the l-th DS. $\Xi_{\mathrm{T}}^{\mathrm{CBR}}$ and $\Xi_{\mathrm{D}_l}^{\mathrm{CBR}}$ are the sets of branches in the TS and the l-th DS. Ξ^{ther} is the set of TUs. $n_{\max}^{\mathrm{T}}$ and $n_{\max}^{\mathrm{D}_l}$ are the maximum numbers of permitted hardened branches in the TS and the l-th DS, respectively. UT_g and DT_g are the minimum start-up/shut-down time of the TU g. s^{TS} denotes the variables determined in the third stage. Y and F are the sets about branch outage probability related to the ambiguity set. These elements will be discussed later.

3.2. Second stage: Natural disaster attacks and the ambiguity set

3.2.1. Modeling of earthquakes

The damage to the power system by an earthquake can be modeled by the amplitude of P-waves or peak ground acceleration. Fig. 5 shows an illustration of the distance between the epicenter of an earthquake and the location of interest. According to [48–49], the peak amplitude of the P-wave to a certain interest can be expressed as

$$\log G_t^{\mathrm{a}}=A_{\mathrm{E}}+2/3\log M_t^{\mathrm{o}}-\log L,\forall t\leqslant T^{\mathrm{SS}} \tag{8a}$$

Here, G_t^{a} represents the peak amplitude of the P-wave, M_t^{o} represents the seismic moment which can be measured by the intensity and energy of the earthquake, and L represents the distance between the hypocenter and certain interests (i.e., the transmission branches and towers). A_{E} is a constant depending on the propagation medium, P-wave velocity, rupture speed, and stress drop of the earthquake [50].

The upper bound of $\overline{G}_t^{\mathrm{a}}$ for the P-wave is calculated as

$$\log\overline{G}_t^{\mathrm{a}}=A_{\mathrm{E}}+2/3\log\overline{M}_t^{\mathrm{o}}-\log L,\forall t\leqslant T^{\mathrm{SS}} \tag{8b}$$

3.2.2. Modeling of the branch outage probability

The power branch fails if any one of its components fails. Besides, the outage probability for branch (i,j) should be calculated with the consideration of the time cumulative effect. That is, at each period, the outage probability should be expressed as

$$\begin{cases}p_{ij,t}^{\mathrm{T}}=1-\prod\limits_{\forall\tau\leqslant t}\left(1-p_{ij,\tau}^{\mathrm{W}}\right),\forall t\leqslant T^{\mathrm{SS}},(i,j)\in\Xi^{\mathrm{CBR}},\\p_{ij,t}^{\mathrm{W}}=1-\prod\limits_{\forall q\in PT_{ij}}\left(1-p_{ij,q,t}^{\mathrm{P}}\left(G_t^{\mathrm{a}}\right)\right),\forall t\leqslant T^{\mathrm{SS}},(i,j)\in\Xi^{\mathrm{CBR}}\\p_{z,t}^{\mathrm{CBR}}=p_{ij,t}^{\mathrm{T}},\forall t\leqslant T^{\mathrm{SS}},z\in\Gamma^{\mathrm{CBR}},(i,j)\in\Xi^{\mathrm{CBR}},z=f^{\mathrm{CBR}}(i,j)\end{cases} \tag{8c}$$

where PT_{ij} is the set of supporting poles of the branch (i,j). $p_{ij,t}^{\mathrm{T}}$ is the outage probability for the branch (i,j) at period t considering the time cumulative effect. $p_{ij,t}^{\mathrm{W}}$ is the outage probability for the branch (i,j) without considering the time cumulative effect. $p_{ij,q,t}^{\mathrm{P}}\left(G_t^{\mathrm{a}}\right)$ is the damage

probability of the pole q supporting the branch (i, j), and it can be modeled as a lognormal cumulative distribution function (CDF) [51]:

$$p^{\mathrm{P}}_{ij,q,t}\left(G^{\mathrm{a}}_{t}\right)=\Phi\left[\ln\left(G^{\mathrm{a}}_{t}/m_{\mathrm{R}}\right)/\xi_{\mathrm{R}}\right],\forall(i,j)\in\Xi^{\mathrm{CBR}},q\in PT_{ij},t\leqslant T^{\mathrm{SS}} \tag{8d}$$

where m_{R} and ξ_{R} are the median and logarithmic standard deviations of the lognormal distribution.

3.2.3. The ambiguity set

The ambiguity set that incorporates both moment and Wasserstein metric information can be expressed as follows [40]:

$$F=\left\{\mathrm{Y}\in Y(\Omega)\left|\begin{array}{c}E_{\mathrm{Y}}\left(1-x_{ij,t}\right)\leqslant\overline{p}_{ij,t},\forall(i,j)\in\Xi^{\mathrm{CBR}},t\leqslant T^{\mathrm{SS}}\\ d_{\mathrm{W}}\left(\mathrm{Y},\mathrm{Y}^{\mathrm{E}}\right)\leqslant\theta\end{array}\right.\right\} \tag{9a}$$

where the support set Ω can be defined as

$$\Omega=\left\{x_{ij,t}\left|\begin{array}{c}\sum\limits_{\forall(i,j)\in\Xi^{\mathrm{CBR}}}x_{ij,t}\geqslant N-k,\forall t\leqslant T^{\mathrm{SS}}\\ x_{ij,t}\leqslant x_{ij,\tau},\forall t\geqslant\tau,\tau\leqslant T^{\mathrm{SS}},(i,j)\in\Xi^{\mathrm{CBR}}\end{array}\right.\right\} \tag{9b}$$

The Wasserstein metric $d_{\mathrm{W}}(\mathrm{Y},\mathrm{Y}^{\mathrm{E}})$ can be defined as

$$d_{\mathrm{W}}\left(\mathrm{Y},\mathrm{Y}^{\mathrm{E}}\right)=\inf_{\Pi}\left\{\int_{x_{ij,t}\in\Omega}\int_{x^{\mathrm{E}}_{ij,t}\in\Omega}\left\|x_{ij,t}-x^{\mathrm{E}}_{ij,t}\right\|^{2}_{2}\Pi\left(dx_{ij,t},dx^{\mathrm{E}}_{ij,t}\right)\right\} \tag{9c}$$

Here, $x_{ij,t}$ represents the branch outage state. $\overline{p}_{ij,t}$ and $x\mathrm{E}\ ij,t$ represent the upper bound of the branch outage probability and branch outage samples. They can be calculated through the historical data of the earthquake parameters [50]. Y and Y^{E} denote true and empirical branch outage probability distributions, respectively. $Y(\Omega)$ represents the set of all probability distributions on a σ-algebra of Ω. θ is the Wasserstein radius representing the distance between Y and Y^{E}, which controls the conservatism of solutions. In this paper, Y^{E} serves as an estimate of Y and can be expressed as $\mathrm{Y}^{\mathrm{E}}=\sum_{s=1}^{N^{\mathrm{s}}}p_s\delta(x^{\mathrm{E}}_{s,ij,t})$, where $x^{\mathrm{E}}_{s,ij,t}$ is the s-th sample of $x^{\mathrm{E}}_{ij,t}$, p_s is the probability of the s-th sample, and $\delta(x^{\mathrm{E}}_{s,ij,t})$ denotes the Dirac measure on $x^{\mathrm{E}}_{s,ij,t}$. N^{s} is the number of samples. In this paper, we assume that the earthquake will result in an N-k contingency, which means no more than k branches can be on outage simultaneously, and the value of k is set to the maximum number of branch outages in all samples. In (9c), Π is the joint distribution of $x_{ij,t}$ and $x^{\mathrm{E}}_{ij,t}$ with marginal distributions Y and Y^{E}.

Generally, Y^{E} converges to Y, such that θ approaches 0 when N^{s} keeps increasing, and θ should satisfy the following expression [40]:

$$\mathrm{P}\left(d_{\mathrm{W}}\left(\mathrm{Y},\mathrm{Y}^{\mathrm{E}}\right)\leqslant\theta\right)=1-\exp\left(-N^{\mathrm{s}}\left(\theta^{2}/C^{2}\right)\right) \tag{9d}$$

and θ can be expressed with a given confidence level η_{c} as

$$\theta=C\left(\frac{1}{N^{\mathrm{s}}}\ln\left(\frac{1}{1-\eta_{\mathrm{c}}}\right)\right)^{1/2} \tag{9e}$$

where C is a constant which can be obtained by applying the method in [52].

3.3. Third Stage: System operations and recovery

TDCS operations and recovery are formulated in the third stage with the consideration of network reconfiguration, repair, and optimal dispatch of UAVs.

1) Objective Function: The objective function should minimize the sum of load shedding cost, MS operation cost, and unit operation cost, yielding that

$$\psi_{\mathrm{TS}}(s)=\min_{s^{\mathrm{FS}}_{\mathrm{p}}}\sup_{\mathrm{Y}\in F}E_{\mathrm{Y}}(\phi_{\mathrm{SS}})$$

$$\phi_{\mathrm{SS}}=\min_{s^{\mathrm{TS}}}\sum_{t\geqslant T^{\mathrm{SS}}}\left(\begin{array}{l}\sum\limits_{g\in\Xi^{\mathrm{ther}}}c^{\mathrm{opr}}_{g}P_{g,t}+\sum\limits_{i\in\Xi^{\mathrm{BS}}}c^{\mathrm{LS}}_{i}k^{\mathrm{LS}}_{i,t}P^{\mathrm{D}}_{i,t}+\sum\limits_{l\in\mathscr{N}_{\mathrm{n}}}\left(\sum\limits_{i\in\delta^{\mathrm{MS}}_{l}}c^{S}_{i}P^{\mathrm{S}}_{i,t}+\sum\limits_{g\in\Xi^{\mathrm{dg}}_{\mathrm{D}l}}c^{\mathrm{opr}}_{g}P_{g,t}\right)\\ +\sum\limits_{l\in\mathscr{N}_{\mathrm{n}}}\left[\begin{array}{l}\kappa^{\mathrm{P}}_{l,t}\left(P^{\mathrm{T}\to\mathrm{D}}_{l,t}-\widetilde{P}^{\mathrm{T}\to\mathrm{D}}_{l,t}\right)+\zeta^{\mathrm{P}}_{l,t}\left(P^{\mathrm{T}\to\mathrm{D}}_{l,t}-\widetilde{P}^{\mathrm{T}\to\mathrm{D}}_{l,t}\right)^{2}+\\ \kappa^{\mathrm{Q}}_{l,t}\left(Q^{\mathrm{T}\to\mathrm{D}}_{l,t}-\widetilde{Q}^{\mathrm{T}\to\mathrm{D}}_{l,t}\right)+\zeta^{\mathrm{Q}}_{l,t}\left(Q^{\mathrm{T}\to\mathrm{D}}_{l,t}-\widetilde{Q}^{\mathrm{T}\to\mathrm{D}}_{l,t}\right)^{2}+\\ \kappa^{\mathrm{P}}_{l,t}\left(P^{\mathrm{D}\to\mathrm{T}}_{l,t}-\widetilde{P}^{\mathrm{D}\to\mathrm{T}}_{l,t}\right)+\zeta^{\mathrm{P}}_{l,t}\left(P^{\mathrm{D}\to\mathrm{T}}_{l,t}-\widetilde{P}^{\mathrm{D}\to\mathrm{T}}_{l,t}\right)^{2}+\\ \kappa^{\mathrm{Q}}_{l,t}\left(Q^{\mathrm{D}\to\mathrm{T}}_{l,t}-\widetilde{Q}^{\mathrm{D}\to\mathrm{T}}_{l,t}\right)+\zeta^{\mathrm{Q}}_{l,t}\left(Q^{\mathrm{D}\to\mathrm{T}}_{l,t}-\widetilde{Q}^{\mathrm{D}\to\mathrm{T}}_{l,t}\right)^{2}\end{array}\right]\\ +\left[\kappa^{\mathrm{U}}_{n+1,t}\left(\mathrm{Z}^{\mathrm{PT}}_{t}-\widetilde{\mathrm{Z}}^{\mathrm{IT}}_{t}\right)+\zeta^{\mathrm{U}}_{n+1,t}\left(\mathrm{Z}^{\mathrm{PT}}_{t}-\widetilde{\mathrm{Z}}^{\mathrm{IT}}_{t}\right)^{2}\right]\\ +\sum\limits_{l\in\mathscr{N}_{\mathrm{n}}}\left[\kappa^{\mathrm{U}}_{l,t}\left(\mathrm{Z}^{\mathrm{PD}_l}_{t}-\widetilde{\mathrm{Z}}^{\mathrm{ID}_l}_{t}\right)+\zeta^{\mathrm{U}}_{l,t}\left(\mathrm{Z}^{\mathrm{PD}_l}_{t}-\widetilde{\mathrm{Z}}^{\mathrm{ID}_l}_{t}\right)^{2}\right]\end{array}\right)$$

$$\mathrm{Z}^{\mathrm{PT}}_{t}=\sum_{\forall j\in\mathscr{N}_{\mathrm{UT}z}}\sum_{z\in\widetilde{W}_{j,t}}\varphi^{\mathrm{P}}_{t}(z),\widetilde{\mathrm{Z}}^{\mathrm{IT}}_{t}=\sum_{\forall j\in\mathscr{N}_{\mathrm{UT}z}}\sum_{z\in\widetilde{W}_{j,t}}\widetilde{\varphi}^{\mathrm{I}}_{t}(z),\forall t\geqslant T^{\mathrm{SS}}$$

$$\mathrm{Z}^{\mathrm{PD}_l}_{t}=\sum_{\forall j\in\mathscr{N}_{\mathrm{UD}_l}}\sum_{z\in\widetilde{W}_{j,t}}\varphi^{\mathrm{P}}_{t}(z),\widetilde{\mathrm{Z}}^{\mathrm{ID}_l}_{t}=\sum_{\forall j\in\mathscr{N}_{\mathrm{UD}_l}}\sum_{z\in\widetilde{W}_{j,t}}\widetilde{\varphi}^{\mathrm{I}}_{t}(z),\forall l\in\mathscr{N}_{\mathrm{n}},\mathrm{t}\geqslant\mathrm{T}^{\mathrm{SS}} \tag{10}$$

where $s^{\mathrm{TS}}=\{P_{g,t}, Q_{g,t}, k^{\mathrm{LS}}_{i.t}, P^{\mathrm{S}}_{i,t}, E^{\mathrm{S}}_{i,t}, x_{ij,t}, s^{\mathrm{flow}}\}$ are the variables determined in the third stage. $s^{\mathrm{flow}}=\{fl^{\mathrm{P}}_{ij,t}, fl^{\mathrm{Q}}_{ij,t}, V_{i,t}, \alpha_{i,t}\}$ are the power flow variables. $P_{g,t}$ and $Q_{g,t}$ are the active and reactive power outputs of TUs in TS or distributed generators (DGs) in DSs. $P^{\mathrm{S}}_{i,t}$ and $E^{\mathrm{S}}_{i,t}$ are the power output and energy storage of the MS i, respectively. $k^{\mathrm{LS}}_{i.t}$ denotes the load shedding ratio at the bus i. $P^{\mathrm{D}}_{i,t}$ is the active power load demand at the bus i. $fl^{\mathrm{P}}_{ij,t}$ and $fl^{\mathrm{Q}}_{ij,t}$ are the active and reactive power flow in the branch (i, j) at period t. $V_{i,t}$ and $\alpha_{i,t}$ denote the voltage magnitude and angle at the bus i. $\kappa^{\mathrm{P}}_{l,t}$, $\kappa^{\mathrm{Q}}_{l,t}$, $\zeta^{\mathrm{P}}_{l,t}$, $\zeta^{\mathrm{Q}}_{l,t}$ are the penalty function multipliers utilized in the transmission and distribution system sub-problems. c^{opr}_{g} is the operation cost for the generation g; c^{LS}_{i} is the penalty cost for shedding loads at the bus i; c^{S}_{i} is the operation cost for the MS i. $\widetilde{P}^{\mathrm{T}\to\mathrm{D}}_{l,t}$ and $\widetilde{Q}^{\mathrm{T}\to\mathrm{D}}_{l,t}$ are the results of the active and reactive power in the l-th contact line solved in the l-th DS sub-problem. $\widetilde{P}^{\mathrm{D}\to\mathrm{T}}_{l,t}$ and $\widetilde{Q}^{\mathrm{D}\to\mathrm{T}}_{l,t}$ are the results of the active and reactive power in the l-th contact lines solved in the TS sub-problem.

The objective function in the third stage can be divided into two parts: the objective function of the TS sub-problem and the objective functions of DS sub-problems, which can be expressed as

$$\psi_{\mathrm{TS}}(s)=\min_{s_{\mathrm{p}}^{\mathrm{FS}}}\sup_{\Upsilon\in F}E_{\Upsilon}\left(\phi_{\mathrm{TS}}^{\mathrm{T}}+\phi_{\mathrm{TS}}^{\mathrm{D}}\right)$$

$$\phi_{\mathrm{TS}}^{\mathrm{T}}=\min_{s_{\mathrm{T}}^{\mathrm{TS}}}\sum_{t\geqslant T^{SS}}\left\{\begin{array}{c}\sum_{g\in\Xi^{\mathrm{ther}}}c_g^{\mathrm{opr}}Pt_{g,t}+\sum_{i\in\Xi_{\mathrm{T}}^{\mathrm{BS}}}c_i^{\mathrm{LS}}k_{i,t}^{\mathrm{TLS}}P_{i,t}^{\mathrm{D}}\\+\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\left[\begin{array}{c}\kappa_{l,t}^{\mathrm{P}}\left(P_{l,t}^{\mathrm{T}\to\mathrm{D}}-\widetilde{P}_{l,t}^{\mathrm{T}\to\mathrm{D}}\right)+\zeta_{l,t}^{\mathrm{P}}\left(P_{l,t}^{\mathrm{T}\to\mathrm{D}}-\widetilde{P}_{l,t}^{\mathrm{T}\to\mathrm{D}}\right)^2+\\\kappa_{l,t}^{\mathrm{Q}}\left(Q_{l,t}^{\mathrm{T}\to\mathrm{D}}-\widetilde{Q}_{l,t}^{\mathrm{T}\to\mathrm{D}}\right)+\zeta_{l,t}^{\mathrm{Q}}\left(Q_{l,t}^{\mathrm{T}\to\mathrm{D}}-\widetilde{Q}_{l,t}^{\mathrm{T}\to\mathrm{D}}\right)^2\end{array}\right]\\+\left[\kappa_{n+1,t}^{\mathrm{U}}\left(\mathrm{Z}_t^{\mathrm{PT}}-\widetilde{\mathrm{Z}}_t^{\mathrm{IT}}\right)+\zeta_{n+1,t}^{\mathrm{U}}\left(\mathrm{Z}_t^{\mathrm{PT}}-\widetilde{\mathrm{Z}}_t^{\mathrm{IT}}\right)^2\right]\end{array}\right\}$$

$$\phi_{\mathrm{TS}}^{\mathrm{D}}=\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\phi_{\mathrm{TS}}^{\mathrm{D}_l}$$

$$\phi_{\mathrm{TS}}^{\mathrm{D}_l}=\min_{s_{\mathrm{D}}^{\mathrm{TS}}}\sum_{t\geqslant T^{SS}}\left[\begin{array}{c}\sum_{g\in\Xi_{\mathrm{D}_l}^{\mathrm{dg}}}c_g^{\mathrm{opr}}Pd_{g,t}+\sum_{i\in\Xi_{\mathrm{D}_l}^{\mathrm{BS}}}c_i^{\mathrm{LS}}k_{i,t}^{\mathrm{DLS}}P_{i,t}^{\mathrm{D}}+\sum_{i\in\delta_l^{\mathrm{MS}}}c_i^{\mathrm{S}}P_{i,t}^{\mathrm{S}}\\\kappa_{l,t}^{\mathrm{P}}\left(P_{l,t}^{\mathrm{D}\to\mathrm{T}}-\widetilde{P}_{l,t}^{\mathrm{D}\to\mathrm{T}}\right)+\zeta_{l,t}^{\mathrm{P}}\left(P_{l,t}^{\mathrm{D}\to\mathrm{T}}-\widetilde{P}_{l,t}^{\mathrm{D}\to\mathrm{T}}\right)^2+\\\kappa_{l,t}^{\mathrm{Q}}\left(Q_{l,t}^{\mathrm{D}\to\mathrm{T}}-\widetilde{Q}_{l,t}^{\mathrm{D}\to\mathrm{T}}\right)+\zeta_{l,t}^{\mathrm{Q}}\left(Q_{l,t}^{\mathrm{D}\to\mathrm{T}}-\widetilde{Q}_{l,t}^{\mathrm{D}\to\mathrm{T}}\right)^2+\\\kappa_{l,t}^{\mathrm{U}}\left(\mathrm{Z}_t^{\mathrm{P}}-\widetilde{\mathrm{Z}}_t^{\mathrm{I}}\right)+\zeta_{l,t}^{\mathrm{U}}\left(\mathrm{Z}_t^{\mathrm{P}}-\widetilde{\mathrm{Z}}_t^{\mathrm{I}}\right)^2\end{array}\right],\forall l\in\mathscr{N}_{\mathrm{n}}\tag{11a}$$

where $\phi_{\mathrm{TS}}^{\mathrm{T}}$ and $\phi_{\mathrm{TS}}^{\mathrm{D}}$ denote the objective functions of the TS sub-problem and DS sub-problems. The variables can be expressed as $s_{\mathrm{T}}^{\mathrm{TS}}=\{Pt_{g,t}, Qt_{g,t}, k_{i,t}^{\mathrm{TLS}}, x_{ij,t}^{\mathrm{T}}, fl_{ij,t}^{\mathrm{TP}}, fl_{ij,t}^{\mathrm{TQ}}, V_{i,t}^{\mathrm{T}}, \alpha_{i,t}^{\mathrm{T}}\}$ and $s_{\mathrm{D}}^{\mathrm{TS}}=\{Pd_{g,t}, Qd_{g,t}, k_{i,t}^{\mathrm{DLS}}, P_{i,t}^{\mathrm{S}}, E_{i,t}^{\mathrm{S}}, x_{ij,t}^{\mathrm{D}}, fl_{ij,t}^{\mathrm{DP}}, fl_{ij,t}^{\mathrm{DQ}}, V_{i,t}^{\mathrm{D}}, \alpha_{i,t}^{\mathrm{D}}\}$. $\Xi_{\mathrm{D}_l}^{\mathrm{dg}}$ is the set of DGs in the l-th DS. $\Xi_{\mathrm{T}}^{\mathrm{BS}}$ and $\Xi_{\mathrm{D}_l}^{\mathrm{BS}}$ are the sets of buses in the TS and the l-th DS. $Pt_{g,t}$ and $Qt_{g,t}$ denote the active and reactive output of the TU g in the TS at each period, respectively. $k_{i,t}^{\mathrm{TLS}}$ denotes the load shedding ratio at the bus i in the TS at each period. $x_{ij,t}^{\mathrm{T}}$ is the outage state of the branch (i, j) in the TS at each period. $fl_{ij,t}^{\mathrm{TP}}$ and $fl_{ij,t}^{\mathrm{TQ}}$ are the active and reactive power flow transmitted in branches in the TS at each period. $V_{i,t}^{\mathrm{T}}$ and $\alpha_{i,t}^{\mathrm{T}}$ are the voltage magnitude and angle at the bus i in the TS at each period. $Pd_{g,t}$ and $Qd_{g,t}$ are the active and reactive output of the DG g in DSs at each period. $k_{i,t}^{\mathrm{DLS}}$ is the load shedding ratio at the bus i in the DSs at each period. $x_{ij,t}^{\mathrm{D}}$ is the outage state of the branch (i, j) in DSs at each period. $fl_{ij,t}^{\mathrm{DP}}$ and $fl_{ij,t}^{\mathrm{DQ}}$ are the active and reactive power flow transmitted in the branch (i, j) in DSs at each period. $V_{i,t}^{\mathrm{T}}$ and $\alpha_{i,t}^{\mathrm{T}}$ are the voltage magnitude and angle at the bus i in DSs at each period.

2) Load Shedding Constraints: The load shedding ratio for power loads should be limited to a certain value. By a proper network reinforcement scheme, the total load shedding under a certain earthquake is also limited by the maximum permissible load shedding Pmax LST and Pmax LSD, yielding that

$$\left\{\begin{array}{c}0\leqslant k_{i,t}^{\mathrm{TLS}}\leqslant\overline{k}_i^{\mathrm{TLS}},\forall i\in\Xi_{\mathrm{T}}^{\mathrm{BS}},t\geqslant T^{\mathrm{SS}}\\\sum_{\forall t\geqslant T^{SS}}\sum_{\forall i\in\Xi_{\mathrm{T}}^{\mathrm{BS}}}k_{i,t}^{\mathrm{TLS}}P_{i,t}^{\mathrm{D}}\leqslant P_{\mathrm{LST}}^{\max}\end{array}\right.\tag{11b}$$

$$\left\{\begin{array}{c}0\leqslant k_{i,t}^{\mathrm{DLS}}\leqslant\overline{k}_i^{\mathrm{DLS}},\forall i\in\Xi_{\mathrm{D}_l}^{\mathrm{BS}},t\geqslant T^{\mathrm{SS}},l\in\mathscr{N}_{\mathrm{n}}\\\sum_{\forall l\in\mathscr{N}_{\mathrm{n}}}\sum_{\forall t\geqslant T^{SS}}\sum_{\forall i\in\Xi_{\mathrm{D}_l}^{\mathrm{BS}}}k_{i,t}^{\mathrm{DLS}}P_{i,t}^{\mathrm{D}}\leqslant P_{\mathrm{LSD}}^{\max}\end{array}\right.\tag{11c}$$

where $\overline{k}_i^{\mathrm{TLS}}$ and $\overline{k}_i^{\mathrm{DLS}}$ are the upper bounds of the load shedding ratio at the bus i in the TS and DSs, respectively.

3) Constraints for the TS: For the TS, the related constraints are modeled in (11d)–(11h). Among them, (11d)–(11f) are defined as power flow constraints. (11g) denotes the constraints about active power output and power ramping for TUs. (11h) denotes the constraints about the limitation of the power transmitted in the contact line between the TS and the l-th DS.

$$\left\{\begin{array}{c}\sum_{g\in\mathscr{G}_i}Pt_{g,t}-\left(1-k_{i,t}^{\mathrm{TLS}}\right)P_{i,t}^{\mathrm{D}}=\sum_{j\in\Lambda_s(i)}fl_{ij,t}^{\mathrm{TP}}-\sum_{j\in\Lambda_e(i)}fl_{ij,t}^{\mathrm{TP}}\\\sum_{g\in\mathscr{G}_i}Qt_{g,t}-\left(1-k_{i,t}^{\mathrm{TLS}}\right)Q_{i,t}^{\mathrm{D}}=\sum_{j\in\Lambda_s(i)}fl_{ij,t}^{\mathrm{TQ}}-\sum_{j\in\Lambda_e(i)}fl_{ij,t}^{\mathrm{TQ}}\\-\pi\leqslant\alpha_{i,t}^{\mathrm{T}}\leqslant\pi,V_i^{\min}\leqslant V_{i,t}^{\mathrm{T}}\leqslant V_i^{\max}\\\forall i\in\Xi_{\mathrm{T}}^{\mathrm{BS}}\backslash\left(\bigcup_{\forall l\in\mathscr{N}_{\mathrm{n}}}\Xi_{\mathrm{T}}^{\mathrm{BTD}_l}\right),t\geqslant T^{\mathrm{SS}}\end{array}\right.\tag{11d}$$

$$\left\{\begin{array}{c}\sum_{g\in\mathscr{G}_i}Pt_{g,t}-\left(1-k_{i,t}^{\mathrm{TLS}}\right)P_{i,t}^{\mathrm{D}}=P_{l,t}^{\mathrm{T}\to\mathrm{D}}+\sum_{j\in\Lambda_s(i)}fl_{ij,t}^{\mathrm{TP}}-\sum_{j\in\Lambda_e(i)}fl_{ij,t}^{\mathrm{TP}}\\\sum_{g\in\mathscr{G}_i}Qt_{g,t}-\left(1-k_{i,t}^{\mathrm{TLS}}\right)Q_{i,t}^{\mathrm{D}}=Q_{l,t}^{\mathrm{T}\to\mathrm{D}}+\sum_{j\in\Lambda_s(i)}fl_{ij,t}^{\mathrm{TQ}}-\sum_{j\in\Lambda_e(i)}fl_{ij,t}^{\mathrm{TQ}}\\-\pi\leqslant\alpha_{i,t}^{\mathrm{T}}\leqslant\pi,V_i^{\min}\leqslant V_{i,t}^{\mathrm{T}}\leqslant V_i^{\max}\\\forall i\in\Xi_{\mathrm{T}}^{\mathrm{BTD}_l},l\in\mathscr{N}_{\mathrm{n}},\mathrm{t}\geqslant\mathrm{T}^{\mathrm{SS}}\end{array}\right.\tag{11e}$$

$$\left\{\begin{array}{c}-M\left(1-x_{ij,t}^{\mathrm{T}}\right)\leqslant fl_{ij,t}^{\mathrm{TP}}\\-\left(G_{ij}V_{j,t}^{\mathrm{T}}-B_{ij}\alpha_{j,t}\right)\leqslant M\left(1-x_{ij,t}^{\mathrm{T}}\right)\\-M\left(1-x_{ij,t}^{\mathrm{T}}\right)\leqslant fl_{ij,t}^{\mathrm{TQ}}\\-\left(B_{ij}V_{j,t}^{\mathrm{T}}-G_{ij}\alpha_{j,t}\right)\leqslant M\left(1-x_{ij,t}^{\mathrm{T}}\right)\\-\overline{fl}_{ij}^{\mathrm{TP}}x_{ij,t}^{\mathrm{T}}\leqslant fl_{ij,t}^{\mathrm{TP}}\leqslant\overline{fl}_{ij}^{\mathrm{TP}}x_{ij,t}^{\mathrm{T}}\\-\overline{fl}_{ij}^{\mathrm{TQ}}x_{ij,t}^{\mathrm{T}}\leqslant fl_{ij,t}^{\mathrm{TQ}}\leqslant\overline{fl}_{ij}^{\mathrm{TQ}}x_{ij,t}^{\mathrm{T}}\end{array}\right.,\forall(i,j)\in\Xi_{\mathrm{T}}^{\mathrm{CBR}},t\geqslant T^{\mathrm{SS}}\tag{11f}$$

$$\left\{\begin{array}{c}w_{g,t}\underline{P}_g^{\mathrm{T}}\leqslant Pt_{g,t}\leqslant w_{g,t}\overline{P}_g^{\mathrm{T}}\\w_{g,t}\underline{Q}_g^{\mathrm{T}}\leqslant Qt_{g,t}\leqslant w_{g,t}\overline{Q}_g^{\mathrm{T}}\\Pt_{g,t}-Pt_{g,t-1}\leqslant RU_g w_{g,t-1}+SU_g u_{g,t}\\Pt_{g,t-1}-Pt_{g,t}\leqslant RD_g w_{g,t}+SD_g v_{g,t}\end{array}\right.,\forall g\in\Xi^{\mathrm{ther}},t\geqslant T^{\mathrm{SS}}\tag{11g}$$

$$\left\{\begin{array}{c}\underline{P}_l^{\mathrm{T}\to\mathrm{D}}\leqslant P_{l,t}^{\mathrm{T}\to\mathrm{D}}\leqslant\overline{P}_l^{\mathrm{T}\to\mathrm{D}}\\\underline{Q}_l^{\mathrm{T}\to\mathrm{D}}\leqslant Q_{l,t}^{\mathrm{T}\to\mathrm{D}}\leqslant\overline{Q}_l^{\mathrm{T}\to\mathrm{D},\max}\end{array}\right.,\forall l\in\mathscr{N}_{\mathrm{n}},\mathrm{t}\geqslant\mathrm{T}^{\mathrm{SS}}\tag{11h}$$

where $P\mathrm{T}\to\mathrm{D}$ l,t, and $Q\mathrm{T}\to\mathrm{D}$ l,t denote the active and reactive power transmitted in the contact line from the TS to the l-th DS. B_{ij} and G_{ij} are the susceptance and conductance in the branch (i, j). RU_g/RD_g is the ramp-up/-down limit of TU g. SU_g/SD_g is the start-up/shut-down ramp limit of TU g. $\underline{P}_l^{\mathrm{T}\to\mathrm{D}}/\overline{P}_l^{\mathrm{T}\to\mathrm{D}}$ and $\underline{Q}_l^{\mathrm{T}\to\mathrm{D}}/\overline{Q}_l^{\mathrm{T}\to\mathrm{D}}$ denote the lower/upper bound of the transmitted active and reactive power in the contact line. $\overline{fl}_{ij}^{\mathrm{TP}}$ and $\overline{fl}_{ij}^{\mathrm{TQ}}$ denote the upper bound of the active and reactive power in the branch (i, j). $\underline{P}_g^{\mathrm{T}}/\overline{P}_g^{\mathrm{T}}$ and $\underline{Q}_g^{\mathrm{T}}/\overline{Q}_g^{\mathrm{T}}$ denote the lower/upper bound of the active and reactive power outputs for the TU g in the TS. $V_j^{\min}$ and $V_j^{\max}$ are the lower and upper bounds of voltage magnitude at the bus j. $\Lambda_s(i)$ and $\Lambda_e(i)$ are the sets of buses with flow inflowing to and outflowing from bus i. $\mathscr{G}_i$ denotes the set of thermal units connected to the bus i. Ξ_T^{BTDl} denotes the set of buses in the TS connected to the contact line between the TS and the l-th DS.

4) Constraints for DSs: (11i) and (11j) denote the power balance constraint at each bus. (11k) is derived as a Distflow formulation. (11l) denotes the upper bound and lower bound of power output for each DG in the DSs. (11m) limits the power transmitted in each contact line

between the TS and DSs. (11n) denotes the constraints on branch state changes. Only the branches covered by UAVs can be reconfigured or repaired.

$$\begin{cases} \sum_{g\in\mathscr{D}_i} Pd_{g,t} - \left(1-k_{i,t}^{\mathrm{DLS}}\right)P_{i,t}^{\mathrm{D}} + \sum_{m\in\mathscr{I}_i} P_{m,t}^{\mathrm{S}} = \sum_{j\in\Lambda_s(i)} fl_{ij,t}^{\mathrm{DP}} - \sum_{j\in\Lambda_e(i)} fl_{ij,t}^{\mathrm{DP}} \\ \sum_{g\in\mathscr{D}_i} Qd_{g,t} - \left(1-k_{i,t}^{\mathrm{DLS}}\right)Q_{i,t}^{\mathrm{D}} = \sum_{j\in\Lambda_s(i)} fl_{ij,t}^{\mathrm{DQ}} - \sum_{j\in\Lambda_e(i)} fl_{ij,t}^{\mathrm{DQ}} \end{cases} \quad (11i)$$
$$\forall i \in \Xi_{\mathrm{D}_l}^{\mathrm{BS}} \backslash \Xi_{\mathrm{D}_l}^{\mathrm{BTD}}, l \in \mathscr{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}}$$

$$\begin{cases} \sum_{g\in\mathscr{D}_i} Pd_{g,t} - \left(1-k_{i,t}^{\mathrm{DLS}}\right)P_{i,t}^{\mathrm{D}} + \sum_{m\in\mathscr{I}_i} P_{m,t}^{\mathrm{S}} = P_{l,t}^{\mathrm{D}\to\mathrm{T}} + \sum_{j\in\Lambda_s(i)} fl_{ij,t}^{\mathrm{DP}} - \sum_{j\in\Lambda_e(i)} fl_{ij,t}^{\mathrm{DP}} \\ \sum_{g\in\mathscr{D}_i} Qd_{g,t} - \left(1-k_{i,t}^{\mathrm{DLS}}\right)Q_{i,t}^{\mathrm{D}} = Q_{l,t}^{\mathrm{D}\to\mathrm{T}} + \sum_{j\in\Lambda_s(i)} fl_{ij,t}^{\mathrm{DQ}} - \sum_{j\in\Lambda_e(i)} fl_{ij,t}^{\mathrm{DQ}} \end{cases}$$
$$\forall i \in \Xi_{\mathrm{D}_l}^{\mathrm{BTD}}, l \in \mathscr{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \quad (11j)$$

$$\begin{cases} -M\left(1-x_{ij,t}^{\mathrm{D}}\right) \leqslant V_{j,t}^{\mathrm{D}} - V_{i,t}^{\mathrm{D}} - \left(Dr_{ij} fl_{ij,t}^{\mathrm{DP}} + Dx_{ij} fl_{ij,t}^{\mathrm{DQ}}\right) \leqslant M\left(1-x_{ij,t}^{\mathrm{D}}\right) \\ \left(fl_{ij,t}^{\mathrm{DP}}\right)^2 + \left(fl_{ij,t}^{\mathrm{DQ}}\right)^2 \leqslant S_{ij}^{\max} x_{ij,t}^{\mathrm{D}} \\ V_j^{\min} \leqslant V_{j,t}^{\mathrm{D}} \leqslant V_j^{\max} \end{cases} \quad (11k)$$
$$\forall (i,j) \in \Xi_{\mathrm{D}_l}^{\mathrm{CBR}}, l \in \mathscr{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}}$$

$$\begin{cases} \underline{P}_g^{\mathrm{D}} \leqslant Pd_{g,t} \leqslant \overline{P}_g^{\mathrm{D}} \\ \underline{Q}_g^{\mathrm{D}} \leqslant Qd_{g,t} \leqslant \overline{Q}_g^{\mathrm{D}} \end{cases}, \forall g \in \Xi_{\mathrm{D}_l}^{\mathrm{dg}}, l \in \mathscr{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \quad (11l)$$

$$\begin{cases} \underline{P}_l^{\mathrm{D}\to\mathrm{T}} \leqslant P_{l,t}^{\mathrm{D}\to\mathrm{T}} \leqslant \overline{P}_l^{\mathrm{D}\to\mathrm{T}} \\ \underline{Q}_l^{\mathrm{D}\to\mathrm{T}} \leqslant Q_{l,t}^{\mathrm{D}\to\mathrm{T}} \leqslant \overline{Q}_l^{\mathrm{D}\to\mathrm{T}} \end{cases}, \forall l \in \mathscr{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \quad (11m)$$

$$\begin{cases} x_{ij,t}^{\mathrm{D}} = x_{ij,t-1}^{\mathrm{D}}, \forall (i,j) \in \Xi_{\mathrm{D}_l}^{\mathrm{CBR}} \backslash \bigcup_{\forall c \in \mathscr{N}_{\mathrm{UD}_l}} \widetilde{W}_{c,t}^{\mathrm{CBR}}, l \in \mathscr{N}_{\mathrm{n}}, \forall \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \\ \widetilde{W}_{j,t}^{\mathrm{CBR}} = \left\{ x \in \Xi^{\mathrm{CBR}} \middle| x = \left(f^{\mathrm{CBR}}\right)^{-1}(y), \forall y \in W_{j,t} \right\}, \forall j \in \mathscr{N}_{\mathrm{U}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \\ \widetilde{W}_{j,t}^{\mathrm{BS}} = \left\{ x \in \Xi^{\mathrm{BS}} \middle| x = \left(f^{\mathrm{BS}}\right)^{-1}(y), \forall y \in W_{j,t} \right\}, \forall j \in \mathscr{N}_{\mathrm{U}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}} \end{cases} \quad (11n)$$

$$\begin{cases} 0 \leqslant P_{i,t}^{\mathrm{S}} \leqslant P_i^{\mathrm{SMax}} \\ E_{i,t}^{\mathrm{S}} = E_{i,0}^{\mathrm{S}} + \sum_{\tau=T^{\mathrm{SS}}}^{t} P_{i,\tau}^{\mathrm{S}}, \ \forall t \geqslant T^{\mathrm{SS}}, i \in \delta_l^{\mathrm{MS}}, l \in \mathscr{N}_{\mathrm{n}} \\ 0 \leqslant E_{i,t}^{\mathrm{S}} \leqslant E_i^{\mathrm{SMax}} \end{cases} \quad (11o)$$

where $P_{l,t}^{\mathrm{D}\to\mathrm{T}}$ and $Q_{l,t}^{\mathrm{D}\to\mathrm{T}}$ denote the active and reactive powers transmitted in the contact line from the l-th DS to the TS. $fl_{ij,t}^{\mathrm{DP}}$ and $fl_{ij,t}^{\mathrm{DQ}}$ are the active and reactive power flow in the branch (i, j) in DSs. Dr_{ij} and Dx_{ij} are the resistance and reactance of the branch (i, j). $S_{ij}^{\max}$ is the capacity of the branch (i, j). $\underline{P}_g^{\mathrm{D}}/\overline{P}_g^{\mathrm{D}}$ and $\underline{Q}_g^{\mathrm{D}}/\overline{Q}_g^{\mathrm{D}}$ denote the lower/upper bound of the active and reactive power outputs for the DG g in the DS. $\underline{P}_l^{\mathrm{D}\to\mathrm{T}}/\overline{P}_l^{\mathrm{D}\to\mathrm{T}}$ and $\underline{Q}_l^{\mathrm{D}\to\mathrm{T}}/\overline{Q}_l^{\mathrm{D}\to\mathrm{T}}$ denote the lower/upper bound of the transmitted active and reactive power in the contact line from the l-th DS to the TS. $\mathscr{D}_i$ denotes the set of DGs connected to the bus i. $\mathscr{I}_i$ denotes the set of MS connected to the bus i. $\Xi_{\mathrm{D}_l}^{\mathrm{BTD}}$ denotes the set of buses in the l-th DS connected to the contact line between the l-th DS and the TS.

The operation of MSs is modeled as (11o). During an outage, MSs will be deployed from the storage center to one charging station to support critical loads, especially those located on an island and lost communication support by UAVs. When the critical loads are restored, MSs will travel to other stations for supporting the additional loads. The MS dispatch can be regarded as a routing problem. Consider that there are N_l charging stations in the l-th DS and the set of charging stations in the l-th DS is GC l= {0, 1, 2, 3, …, N_l, *end*}, where "0" and "*end*" denote the storage center. The dispatch of MS should follow three principles: (i) Each MS should start from the storage center and return to it after the process is completed. Also, MSs that come to a charging station must leave when charging is completed. (ii) For one charging station, several different MSs are allowed for each outage, and the charging time for each MS can be variable. (iii) For each MS, the arriving time for this MS at one charging station equals the leaving time for this MS at the last charging station plus the travel time between these two charging stations. To mathematically describe the above operation principles, (11o) is proposed as follows [53].

$$\begin{cases} \sum_{n\in G_l^{\mathrm{C}}} M_{o,n,i}^{\mathrm{ms}} - \sum_{m\in G_l^{\mathrm{C}}} M_{m,0,i}^{\mathrm{ms}} = 1 \\ \sum_{m\in G_l^{\mathrm{C}}} M_{m,end,i}^{\mathrm{ms}} - \sum_{n\in G_l^{\mathrm{C}}} M_{end,n,i}^{\mathrm{ms}} = 1 \\ \sum_{n\in G_l^{\mathrm{C}}} M_{s,n,i}^{\mathrm{ms}} - \sum_{m\in G_l^{\mathrm{C}}} M_{m,s,i}^{\mathrm{ms}} = 0, \forall s \in G_l^{\mathrm{C}} \\ T_{n,i}^{\mathrm{ms}} = T_{m,i}^{\mathrm{ms}} + t_{m,n,i}^{\mathrm{travel}} + t_{m,i}^{\mathrm{ms}}, \forall (m,n) \in G_l^{\mathrm{C}} \\ -\left(1-M_{m,n,i}^{\mathrm{ms}}\right)M \leqslant T_{m,i}^{\mathrm{ms}} + t_{m,i}^{\mathrm{ms}} + t_{m,n}^{\mathrm{travel}} - T_{n,i}^{\mathrm{ms}} \\ \quad \leqslant \left(1-M_{m,n,i}^{\mathrm{ms}}\right)M, \forall (m,n) \in G_l^{\mathrm{C}} \\ T_{0,i}^{\mathrm{ms}} = 0, \sum_{\forall t} fc_{m,t,i} \leqslant 1, \forall m \in G_l^{\mathrm{C}} \\ \sum_{\forall t} fc_{m,t,i} = \sum_{\forall t} lc_{m,t,i}, \forall m \in G_l^{\mathrm{C}} \\ T_{m,i}^{\mathrm{ms}} \leqslant \sum_{\forall t} \left(t \bullet fc_{m,t,i}\right) \leqslant T_{m,i}^{\mathrm{ms}} + 1 - \varepsilon, \forall m \in G_l^{\mathrm{C}} \\ \sum_{\forall t} \left(t \bullet lc_{m,t,i}\right) \geqslant T_{m,i}^{\mathrm{ms}} + t_{m,i}^{\mathrm{ms}}, \forall m \in G_l^{\mathrm{C}} \\ \sum_{\forall t} \left(t \bullet lc_{m,t,i}\right) \leqslant T_{m,i}^{\mathrm{ms}} + t_{m,i}^{\mathrm{ms}} + 1 - \varepsilon, \forall m \in G_l^{\mathrm{C}} \\ t_{i,d}^{\mathrm{ms}} \geqslant 0, \forall m \in G_l^{\mathrm{C}}, \forall d \\ 0 \leqslant T_{i,d}^{\mathrm{ms}} \leqslant M \sum_{m\in G_l^{\mathrm{C}}} M_{m,n,i}^{\mathrm{ms}}, \forall n \in G_l^{\mathrm{C}} \\ con_{m,t,i} = \sum_{\tau=1}^{t-1} \left(fc_{m,\tau,i} - lc_{m,\tau,i}\right), \forall m \in G_l^{\mathrm{C}} \\ \sum_{\forall m\in G_l^{\mathrm{C}}} con_{m,t,i} \leqslant 1 \\ 0 \leqslant P_{m,t,i}^{\mathrm{S}} \leqslant P_i^{\mathrm{SMax}} con_{m,t,i}, \forall m \in G_l^{\mathrm{C}} \\ E_{t,i}^{\mathrm{S}} = E_{o,i}^{\mathrm{S}} + \sum_{\tau=T^{\mathrm{SS}}}^{t} \sum_{\forall m\in G_l^{\mathrm{C}}} P_{m,\tau,i}^{\mathrm{S}} \\ 0 \leqslant E_{t,i}^{\mathrm{S}} \leqslant E_{t,i}^{\mathrm{SMax}} \end{cases}, \ \forall t \geqslant T^{\mathrm{SS}}, i \in \delta_l^{\mathrm{MS}}, l \in \mathscr{N}_{\mathrm{n}} \quad (11p)$$

where $M_{m,n,i}^{\mathrm{ms}}$ is a binary variable indicating whether the MS i travels from charging station m to charging station n. $T_{m,i}^{\mathrm{ms}}$ denotes when the MS i arrives at charging station m. $t_{m,n,i}^{\mathrm{travel}}$ is the travel time for the MS i between charging stations m and n. $t_{m,i}^{\mathrm{ms}}$ is the charging time for the MS i at charging station m. $fc_{m,t,i}$ and $lc_{m,t,i}$ are two auxiliary binary variables that are utilized to indicate when the MS i arrives at and leaves from the charging station i. For example, for a certain charging station m and the related MS i which is meant to finish the charging task at station m, if the MS i arrives at station m at the time period τ, $fc_{m,\tau,i} = 1$. For $t \neq \tau$, $fc_{m,t,i} =$

0. Similarly, $lc_{m,t,i}$ is defined in the same method. If the MS i leaves from station m at time period τ, $fc_{m,\tau,i} = 1$. For $t \neq \tau$, $lc_{m,t,i} = 0$. $con_{m,t,i}$ is a binary variable indicating whether the MS i implements the charging task in the station m at time period t. GC $l$is a set that contains all charging stations in the l-th DS. M is a constant that is utilized in the big-M approach.

5) Constraints for the Coordination of the TDCS: The power transmitted to each system from the contact lines can be regarded as virtual power loads or virtual power generations. By taking the equivalent of virtual power generations or virtual loads, the collaborative planning and operation problem can be decomposed into multiple sub-problems within the TS and DSs. The power transmitted in contact lines should follow (11p).

$$\begin{cases} \boldsymbol{\varpi}_{l,t} - \boldsymbol{\vartheta}_{l,t} = \mathbf{0} \\ \boldsymbol{\varpi}_{l,t} = \left[P_{l,t}^{\mathrm{T}\to\mathrm{D}}, Q_{l,t}^{\mathrm{T}\to\mathrm{D}}\right] \\ \boldsymbol{\vartheta}_{l,t} = \left[P_{l,t}^{\mathrm{D}\to\mathrm{T}}, Q_{l,t}^{\mathrm{D}\to\mathrm{T}}\right] \end{cases}, \forall t \geqslant T^{\mathrm{SS}}, l \in \mathscr{N}_{\mathrm{n}} \tag{11q}$$

where $\boldsymbol{\varpi}_{l,t}$ and $\boldsymbol{\vartheta}_{l,t}$ are the equivalent virtual power loads in the TS and virtual power generations in DSs.

6) Constraints for Network Repairing and Reconfiguration.

After the earthquake, the branches which lie in coverage zones of UAVs and regain communication can access the repair plans. Constraints (11q) and (11r) give the disconnected state of branches in the TS and DSs. Constraints (11s) and (11t) indicate the state changing for each branch. (11t) indicates that faulted branches in DSs can be reconfigured after they are repaired.

$$\begin{cases} x_{ij,t}^{\mathrm{T}} = 0, \forall (i,j) \in \mathbb{W}_{\mathrm{t}}^{\mathrm{T}} \\ \mathbb{U}_{\mathrm{t}}^{\mathrm{T}} = \mathbb{W}_{\mathrm{t}}^{\mathrm{T}} \cap \left(\bigcup_{\forall \mathrm{j} \in \mathscr{N}_{\mathrm{UT}}} \widetilde{\mathrm{W}}_{\mathrm{j,t}}^{\mathrm{CBR}} \right) \end{cases}, \forall t \geqslant T^{\mathrm{SS}} \tag{11r}$$

$$\begin{cases} x_{ij,t}^{\mathrm{D}} = 0, \forall (i,j) \in \mathbb{W}_{\mathrm{t}}^{\mathrm{D_l}} \\ \mathbb{U}_{\mathrm{t}}^{\mathrm{D_l}} = \mathbb{W}_{\mathrm{t}}^{\mathrm{D_l}} \cap \left(\bigcup_{\forall \mathrm{j} \in \mathscr{N}_{\mathrm{UD_l}}} \widetilde{\mathrm{W}}_{\mathrm{j,t}}^{\mathrm{CBR}} \right) \end{cases}, \forall t \geqslant T^{\mathrm{SS}}, l \in \mathscr{N}_{\mathrm{n}} \tag{11s}$$

$$\begin{cases} \xi_{ij,t+1} - \xi_{ij,t} \leqslant \xi_{ij,t+t_r}, \ T^{\mathrm{SS}} \leqslant t \leqslant T^{\mathrm{SS}} - t_r \\ \xi_{ij,t+t_r} + \xi_{ij,t} \leqslant 1, \ T^{\mathrm{SS}} \leqslant t \leqslant T^{\mathrm{SS}} - t_r \\ \sum \xi_{ij,t} \leqslant \xi_t^{T,max}, \forall t \geqslant T^{\mathrm{SS}} \\ -M \sum_{\tau=t}^{t+t_r-1} \left(1 - \xi_{ij,\tau}\right) \leqslant fic_{ij,t} - 1 \leqslant M \sum_{\tau=t}^{t+t_r-1} \left(1 - \xi_{ij,\tau}\right), \forall t \geqslant T^{\mathrm{SS}} \\ -M\left(1 - fic_{ij,t+1}\right) \leqslant \xi_{ij,t} - 1 \leqslant M\left(1 - fic_{ij,t+1}\right), \forall t \geqslant T^{\mathrm{SS}} \\ \sum_{\forall t \geqslant T^{\mathrm{SS}}} fic_{ij,t} = 1 \\ x_{ij,t}^{\mathrm{T}} = \sum_{\tau=1}^{t} fic_{ij,\tau} \end{cases} \tag{11t}$$

$$\forall (i,j) \in \mathbb{U}_{\mathrm{t}}^{\mathrm{T}}$$

$$\begin{cases} \xi_{ij,t+1} - \xi_{ij,t} \leqslant \xi_{ij,t+t_r}, \ T^{\mathrm{SS}} \leqslant t \leqslant T^{\mathrm{SS}} - t_r \\ \xi_{ij,t+t_r} + \xi_{ij,t} \leqslant 1, \ T^{\mathrm{SS}} \leqslant t \leqslant T^{\mathrm{SS}} - t_r \\ \sum \xi_{ij,t} \leqslant \xi_t^{D_l,\max}, \forall t \geqslant T^{\mathrm{SS}} \\ -M \sum_{\tau=t}^{t+t_r-1} \left(1 - \xi_{ij,\tau}\right) \leqslant fic_{ij,t} - 1 \leqslant M \sum_{\tau=t}^{t+t_r-1} \left(1 - \xi_{ij,\tau}\right), \forall t \geqslant T^{\mathrm{SS}} \\ -M\left(1 - fic_{ij,t+1}\right) \leqslant \xi_{ij,t} - 1 \leqslant M\left(1 - fic_{ij,t+1}\right), \forall t \geqslant T^{\mathrm{SS}} \\ \sum_{\forall t \geqslant T^{\mathrm{SS}}} fic_{ij,t} = 1 \\ x_{ij,t}^{\mathrm{D}} \leqslant \sum_{\tau=1}^{t} fic_{ij,\tau}, \forall t \geqslant T^{\mathrm{SS}} \end{cases} \tag{11u}$$

$$\forall (i,j) \in \mathbb{U}_{\mathrm{t}}^{\mathrm{D_l}}, l \in \mathscr{N}_{\mathrm{n}}$$

where $\mathbb{W}_t$ is the set of faulted branches at each period. $\mathbb{U}_t^{\mathrm{T}}$ and $\mathbb{U}_t^{\mathrm{D_l}}$ are the sets of faulted branches that lie on the coverage zones of UAVs in the TS and the l-the DS. $\xi_{ij,t}$ is the repair state of the branch (i, j): $\xi_{ij,t} = 1$ if repairing; 0, otherwise. $\xi_t^{T,max}$ and $\xi_t^{D_l,\max}$ are the maximum numbers of the branches allowed to be repaired in the TS and the l-th DS at period t. t_r is the repair time for branches. $fic_{ij,t}$ is an auxiliary variable indicating when the branch (i, j) is repaired: 1 if completely repaired; 0, at other times.

4. Proposed solution algorithm

We begin this section by presenting the model in vector/matrix forms. Then, we derive the reformulation of the physical part of MPDRRM with the strengthened ambiguity set. Finally, we use a modified ATC algorithm embedded with nested CCG algorithms to solve the proposed model. Before this section, we have to declare that for any matrix A, the transpose of A is represented as A^T.

4.1. Compact formulation

The complete physical part of the MPDRRM model proposed in section III can be expressed as

$$\begin{aligned} \psi(s) &= \psi_{\mathrm{FS}}(s) + \psi_{\mathrm{TS}}(s) \\ &= \min_{s_{\mathrm{u}}^{\mathrm{FS}}} \sum_{\forall t \geqslant T^{\mathrm{SS}}} \sum_{g \in \Xi^{\mathrm{ther}}} \left(c_g^{\mathrm{SU}} u_{g,t} + c_g^{\mathrm{SD}} v_{g,t} + c_g^{\mathrm{NL}} w_{g,t} \right) \\ &\quad + \min_{s_{\mathrm{p}}^{\mathrm{FS}}} \sup_{\mathrm{Y} \in F} E_{\mathrm{Y}}\left(\phi_{\mathrm{FS}} + \phi_{\mathrm{SS}}^{\mathrm{T}} + \phi_{\mathrm{SS}}^{\mathrm{D}}\right) \end{aligned} \tag{12a}$$

(7b)–(7h), (11b)–(11s) (12b).

Mathematically, the physical part of the MPDRRM can be expressed as a two-stage distributionally robust programming. For notational brevity, the physical part of the MPDRRM model (12) can also be expressed in vectors and matrices as

$$\min_{\boldsymbol{x}} \left(\boldsymbol{c}^T \boldsymbol{x} + \sup_{\mathrm{Y} \in F} E_{\mathrm{Y}}(Q(\boldsymbol{x}, \boldsymbol{z})) \right) \tag{13a}$$

$$s.t. \ \boldsymbol{A}\boldsymbol{x} \leqslant \boldsymbol{b} \tag{13b}$$

where

$$Q(\boldsymbol{x},z)=\min_{\boldsymbol{y}}\boldsymbol{d}^T\boldsymbol{y} \tag{13c}$$

$$s.t.\ \boldsymbol{G}\boldsymbol{x}+\boldsymbol{M}\boldsymbol{y}+\boldsymbol{N}z\leqslant\boldsymbol{h} \tag{13d}$$

Herein, (7a), (7b)–(7h), (11a), and (11b)–(11t) are represented in (13a), (13b), (13c), and (13d), respectively. The ambiguity set F in (9a) and the related support set Ω in (9b) are also represented as

$$F=\left\{\mathrm{Y}\in Y(\Omega)\left|\begin{array}{l}E_{\mathrm{Y}}(\mathbf{1}-z)\leqslant\boldsymbol{p}\\ d_{\mathrm{W}}\left(\mathrm{Y},\mathrm{Y}^{\mathrm{E}}\right)\leqslant\theta\end{array}\right.\right\} \tag{13e}$$

$$\Omega=\{z|\boldsymbol{I}z\leqslant\boldsymbol{J}\} \tag{13f}$$

The proposed model (13) can be separated into the TS sub-problem (14) and DS sub-problems (15).

$$\min_{\boldsymbol{x}_T}\left(\boldsymbol{c}^T\boldsymbol{x}_{\mathrm{T}}+\sup_{\mathrm{Y}\in F}E_{\mathrm{Y}}(Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}}))\right) \tag{14a}$$

$$s.t.\ \boldsymbol{A}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T}}\leqslant\boldsymbol{b}_{\mathrm{T}} \tag{14b}$$

$$Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})=\min_{\boldsymbol{y}_{\mathrm{T}}}(\boldsymbol{d}_{\mathrm{T}})^T\boldsymbol{y}_{\mathrm{T}} \tag{14c}$$

$$\boldsymbol{E}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T}}+\boldsymbol{G}_{\mathrm{T}}\boldsymbol{y}_{\mathrm{T}}+\boldsymbol{M}_{\mathrm{T}}z_{\mathrm{T}}\leqslant\boldsymbol{h}_{\mathrm{T}} \tag{14d}$$

$$\min_{\boldsymbol{x}_{\mathrm{D}}}\left(\sup_{\mathrm{Y}\in F}E_{\mathrm{Y}}(Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}}))\right) \tag{15a}$$

$$s.t.\ \boldsymbol{A}_{\mathrm{D}}\boldsymbol{x}_{\mathrm{D}}\leqslant\boldsymbol{b}_{\mathrm{D}} \tag{15b}$$

$$Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})=\min_{\boldsymbol{y}_{\mathrm{D}}}(\boldsymbol{d}_{\mathrm{D}})^T\boldsymbol{y}_{\mathrm{D}} \tag{15c}$$

$$\boldsymbol{E}_{\mathrm{D}}\boldsymbol{x}_{\mathrm{D}}+\boldsymbol{G}_{\mathrm{D}}\boldsymbol{y}_{\mathrm{D}}+\boldsymbol{M}_{\mathrm{D}}z_{\mathrm{D}}\leqslant\boldsymbol{h}_{\mathrm{D}} \tag{15d}$$

The function ϕ_{GEN} in (7a) is represented in (14a). (7d), (7h)–(7f) are represented in (14b). The functions $\phi_{\mathrm{FS}}^{\mathrm{T}}$ in (7a) and $\phi_{\mathrm{TS}}^{\mathrm{T}}$ in (11a) are represented in (14c). (11b), (11d)–(11h), (11p)–(11q), and (11s) are represented in (14d). (7b)–(7c), and (7e) are represented in (15b). The functions $\phi_{\mathrm{FS}}^{\mathrm{D}}$ in (7a) and $\phi_{\mathrm{TS}}^{\mathrm{D}}$ in (11a) are represented in (15c). (11c), (11i)–(11p), (11r), and (11t) are represented in (15d).

4.2. Modified nested CCG algorithm

We first rewrite the Wasserstein metric (9c) as

$$d_{\mathrm{W}}\left(\mathrm{Y},\mathrm{Y}^{\mathrm{E}}\right)=\inf_{\mathrm{Y}_s}\left\{\sum_{\forall s\in N^{\mathrm{S}}}\int_{z\in\Omega}p_s\left\|z-z_s^{\mathrm{E}}\right\|_2^2\mathrm{Y}_s(dz)\right\} \tag{16a}$$

$$\Pi\left(dz,dz^{\mathrm{E}}\right)=p_s\mathrm{Y}_s(dz) \tag{16b}$$

where $\mathrm{Y}_s(d\mathbf{z})$ is the conditional distribution of Y when $\mathbf{z}^{\mathrm{E}}=\mathbf{z}_{\mathbf{s}}{}^{\mathrm{E}}$ [54]. Then, the supremum terms in (14) and (15) can be rewritten as

$$\begin{aligned}\sup_{\mathrm{Y}\in F}E_{\mathrm{Y}}(Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}}))&=\max_{\mathrm{Y}}\int_{z_{\mathrm{T}}\in\Omega}Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})\mathrm{Y}(dz_{\mathrm{T}})\\&=\max_{\mathrm{Y}_s}\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{T}}\in\Omega}p_sQ_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})\mathrm{Y}_s(dz_{\mathrm{T}})\end{aligned} \tag{16c}$$

$$\begin{aligned}\sup_{\mathrm{Y}\in F}E_{\mathrm{Y}}(Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}}))&=\max_{\mathrm{Y}}\int_{z_{\mathrm{D}}\in\Omega}Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})\mathrm{Y}(dz_{\mathrm{D}})\\&=\max_{\mathrm{Y}_s}\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{D}}\in\Omega}p_sQ_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})\mathrm{Y}_s(dz_{\mathrm{D}})\end{aligned} \tag{16d}$$

The law of total probability can be expressed as (16e), and the moment constraint in (13e) can be rewritten into (16f) by employing (16e).

$$\mathrm{Y}(dz)=\sum_{\forall s\in N^{\mathrm{S}}}p_s\mathrm{Y}_s(dz) \tag{16e}$$

$$\begin{aligned}E_{\mathrm{Y}}(\mathbf{1}-z)&=\int_{z\in\Omega}(\mathbf{1}-z)\mathrm{Y}(dz)\\&=\sum_{\forall s\in N^{\mathrm{S}}}\int_{z\in\Omega}p_s(\mathbf{1}-z)\mathrm{Y}_s(dz)\end{aligned} \tag{16f}$$

The second-stage supremum terms in (16c) and (16d) with $\mathbf{z}$ subject to the ambiguity set F proposed in (13e) can be transformed into the semi-infinite formulation with dual variables $\rho_{\mathrm{T},s}$, $\rho_{\mathrm{D},s}$, $\boldsymbol{\beta}_{\mathrm{T}}$, $\boldsymbol{\beta}_{\mathrm{D}}$, γ_{T}, γ_{D} as

$$\begin{aligned}&\max_{\mathrm{Y}_s}\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{T}}\in\Omega}p_sQ_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})\mathrm{Y}_s(dz_{\mathrm{T}})\\&\quad s.t.\ \int_{z_{\mathrm{T}}\in\Omega}\mathrm{Y}_s(dz_{\mathrm{T}})=1\ :\rho_{\mathrm{T},s}\\&\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{T}}\in\Omega}p_s(\mathbf{1}-z_{\mathrm{T}})\mathrm{Y}_s(dz_{\mathrm{T}})\leqslant\boldsymbol{p}_{\mathrm{T}}\ :\boldsymbol{\beta}_{\mathrm{T}}\\&\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{T}}\in\Omega}p_s\left\|z_{\mathrm{T}}-z_{\mathrm{T},s}^{\mathrm{E}}\right\|_2^2\mathrm{Y}_s(dz_{\mathrm{T}})\leqslant\theta_{\mathrm{T}}\ :\gamma_{\mathrm{T}}\end{aligned} \tag{16g}$$

$$\begin{aligned}&\max_{\mathrm{Y}_s}\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{D}}\in\Omega}p_sQ_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})\mathrm{Y}_s(dz_{\mathrm{D}})\\&\quad s.t.\ \int_{z_{\mathrm{D}}\in\Omega}\mathrm{Y}_s(dz_{\mathrm{D}})=1\ :\rho_{\mathrm{D},s}\\&\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{D}}\in\Omega}p_s(\mathbf{1}-z_{\mathrm{D}})\mathrm{Y}_s(dz_{\mathrm{D}})\leqslant\boldsymbol{p}\ :\boldsymbol{\beta}_{\mathrm{D}}\\&\sum_{\forall s\in N^{\mathrm{S}}}\int_{z_{\mathrm{D}}\in\Omega}p_s\left\|z_{\mathrm{D}}-z_{\mathrm{D},s}^{\mathrm{E}}\right\|_2^2\mathrm{Y}_s(dz_{\mathrm{D}})\leqslant\theta\ :\gamma_{\mathrm{D}}\end{aligned} \tag{16h}$$

By applying Slater's condition and further reformulating, (16g) and (16h) can be derived into the following reformulated TS sub-problem (17).

$$\begin{aligned}\min_{\boldsymbol{x}_{\mathrm{T}},\rho_{\mathrm{T},s},\boldsymbol{\beta}_{\mathrm{T}}\geqslant0,\gamma_{\mathrm{T}}\geqslant0,\boldsymbol{\varepsilon}_{\mathrm{T},s}}&\boldsymbol{c}^T\boldsymbol{x}_{\mathrm{T}}+\sum_{\forall s\in N^{\mathrm{S}}}p_s\rho_{\mathrm{T},s}+(\boldsymbol{p}_{\mathrm{T}}-\mathbf{1})^T\boldsymbol{\beta}_{\mathrm{T}}\ +\ \theta_{\mathrm{T}}\gamma_{\mathrm{T}}\\&s.t.\ \boldsymbol{A}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T}}\leqslant\boldsymbol{b}_{\mathrm{T}}\\&\left\|\boldsymbol{\varepsilon}_{\mathrm{T},s}\right\|_2^2\leqslant\gamma_{\mathrm{T}}\\\rho_{\mathrm{T},s}\geqslant\max_{z_{\mathrm{T}}\in\Omega}&\left[Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})+(\boldsymbol{\beta}_{\mathrm{T}})^Tz_{\mathrm{T}}-\left(\boldsymbol{\varepsilon}_{\mathrm{T},s}\right)^T\left(z_{\mathrm{T}}-z_{\mathrm{T},s}^{\mathrm{E}}\right)\right]\end{aligned} \tag{17a}$$

where

$$Q_{\mathrm{T}}(\boldsymbol{x}_{\mathrm{T}},z_{\mathrm{T}})=\min_{\boldsymbol{y}_{\mathrm{T}}}(\boldsymbol{d}_{\mathrm{T}})^T\boldsymbol{y}_{\mathrm{T}} \tag{17b}$$

$$s.t.\ \boldsymbol{E}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T}}+\boldsymbol{G}_{\mathrm{T}}\boldsymbol{y}_{\mathrm{T}}+\boldsymbol{M}_{\mathrm{T}}z_{\mathrm{T}}\leqslant\boldsymbol{h}_{\mathrm{T}} \tag{17c}$$

and the following DS sub-problems (18)

$$\begin{aligned}\min_{\boldsymbol{x}_{\mathrm{D}},\rho_{\mathrm{D},s},\boldsymbol{\beta}_{\mathrm{D}}\geqslant0,\gamma_{\mathrm{D}}\geqslant0,\boldsymbol{\varepsilon}_{\mathrm{D},s},\forall s\in N^{\mathrm{S}}}&\sum p_s\rho_{\mathrm{D},s}+(\boldsymbol{p}_{\mathrm{D}}-\mathbf{1})^T\boldsymbol{\beta}_{\mathrm{D}}\ +\ \theta_{\mathrm{D}}\gamma_{\mathrm{D}}\\&s.t.\ \boldsymbol{A}_{\mathrm{D}}\boldsymbol{x}_{\mathrm{D}}\leqslant\boldsymbol{b}_{\mathrm{D}}\\&\left\|\boldsymbol{\varepsilon}_{\mathrm{D},s}\right\|_2^2\leqslant\gamma_{\mathrm{D}}\\\rho_{\mathrm{D},s}\geqslant\max_{z_{\mathrm{D}}\in\Omega}&\left[Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})+(\boldsymbol{\beta}_{\mathrm{D}})^Tz_{\mathrm{D}}-\left(\boldsymbol{\varepsilon}_{\mathrm{D},s}\right)^T\left(z_{\mathrm{D}}-z_{\mathrm{D},s}^{\mathrm{E}}\right)\right]\end{aligned} \tag{18a}$$

where

$$Q_{\mathrm{D}}(\boldsymbol{x}_{\mathrm{D}},z_{\mathrm{D}})=\min_{\boldsymbol{y}_{\mathrm{D}}}(\boldsymbol{d}_{\mathrm{D}})^T\boldsymbol{y}_{\mathrm{D}} \tag{18b}$$

$$s.t.\ \boldsymbol{E}_{\mathrm{D}}\boldsymbol{x}_{\mathrm{D}}+\boldsymbol{G}_{\mathrm{D}}\boldsymbol{y}_{\mathrm{D}}+\boldsymbol{M}_{\mathrm{D}}z_{\mathrm{D}}\leqslant\boldsymbol{h}_{\mathrm{D}} \tag{18c}$$

Thus, (17b)–(17c) and (18b)–(18c) can be equivalently converted into dual formulations [41]:

Table 1
Flowchart of the proposed AIF method.

OCL algorithm for the UAV optimal dispatch	
Step O-1	Give the preset convergence precision ε and set $k \leftarrow 0$, $\eta = +\infty$, $\sigma_j(0) \leftarrow (0, 0)$, $\forall j \in \mathcal{N}_n$;
Step O-2	**if** η is smaller than ε
Step O-3	**Stop**;
Step O-4	**else**
Step O-5	go to **Step O-7**;
Step O-6	**End**
Step O-7	Get $CV_j(Vor_j(\sigma_j(k)))$, $\forall j \in \mathcal{N}_U$ by solving (4d);
Step O-8	Giving $\sigma_j(k+1) = CV_j(Vor_j(\sigma_j(k)))$, $\forall j \in \mathcal{N}_U$, giving $\eta = \sum_{\forall j \in \mathcal{N}_U} \|\sigma_j(k+1) - \sigma_j(k)\|_2^2$;
Step O-9	Set $k \leftarrow k+1$ and go to **Step O-2**;

$$Q_T(\boldsymbol{x}_T, z_T) = \max_{\boldsymbol{\omega}_T \leqslant 0} (\boldsymbol{h}_T - \boldsymbol{E}_T \boldsymbol{x}_T - \boldsymbol{M}_T z_T)^T \boldsymbol{\omega}_T \tag{19a}$$

$$s.t.\ (\boldsymbol{G}_T)^T \boldsymbol{\omega}_T = \boldsymbol{d}_T \tag{19b}$$

$$Q_D(\boldsymbol{x}_D, z_D) = \max_{\boldsymbol{\omega}_D \leqslant 0} (\boldsymbol{h}_D - \boldsymbol{E}_D \boldsymbol{x}_D - \boldsymbol{M}_D z_D)^T \boldsymbol{\omega}_D \tag{19c}$$

$$s.t.\ (\boldsymbol{G}_D)^T \boldsymbol{\omega}_D = \boldsymbol{d}_D \tag{19d}$$

where $\boldsymbol{\omega}_T$ and $\boldsymbol{\omega}_D$ are the dual variables of (17c) and (18c).

The procedure of the modified nested CCG algorithm for the TS sub-problem (17) is detailed as follows [41].

Step T-0: Specify an initial branch outage set $Z_T = \{\mathbf{z}_{T,0,s}\}$ and a convergence tolerance μ_T. Set the lower bound LB to -∞, the upper bound UB to +∞, and the iteration counter $c = 1$.

Step T-1: Solve the following master problem.

$$\begin{aligned} \min_{\boldsymbol{x}_T, \rho_{T,s}, \boldsymbol{\beta}_T \geqslant \mathbf{0}, \gamma_T \geqslant 0, \boldsymbol{\varepsilon}_{T,s}} \ & \boldsymbol{c}^T \boldsymbol{x}_T + \sum_{\forall s \in N^s} p_s \rho_{T,s} + (\boldsymbol{p}_T - \mathbf{1})^T \boldsymbol{\beta}_T + \theta_T \gamma_T \\ & s.t.\ \boldsymbol{A}_T \boldsymbol{x}_T \leqslant \boldsymbol{b}_T \\ & \|\boldsymbol{\varepsilon}_{T,s}\|_2^2 \leqslant \gamma_T \\ \rho_{T,s} \geqslant (\boldsymbol{d}_T)^T \boldsymbol{y}_{T,l,s} + (\boldsymbol{\beta}_T)^T \boldsymbol{z}_{T,l,s} - (\boldsymbol{\varepsilon}_{T,s})^T & \left(z_{T,l,s} - z_{T,s}^E\right), \forall l < c \\ \boldsymbol{E}_T \boldsymbol{x}_T + \boldsymbol{G}_T \boldsymbol{y}_{T,l,s} + \boldsymbol{M}_T \boldsymbol{z}_{T,l,s} & \leqslant \boldsymbol{h}_T, \forall l < c \end{aligned} \tag{20a}$$

Then, let ($\mathbf{x}_{T,c}$, $\rho_{T,c,s}$, $\boldsymbol{\beta}_{T,c}$, $\gamma_{T,c}$, $\boldsymbol{\varepsilon}_{T,c,s}$) and ϕ_c^{MP} be the solution and objective value of the master problem, and update the lower bound $LB = \phi_c^{MP}$.

Step T-2: Solve the corresponding N^s sub-problems based on the optimal solution of the master problem derived in Step T-1. The s-th sub-problem is given as follows.

$$\begin{aligned} \max_{\boldsymbol{\omega}_T \leqslant 0} (\boldsymbol{h}_T - \boldsymbol{E}_T \boldsymbol{x}_T - \boldsymbol{M}_T z_T)^T \boldsymbol{\omega}_T + & \left(\boldsymbol{\beta}_{T,c}\right)^T \boldsymbol{z}_T - \left(\boldsymbol{\varepsilon}_{T,c,s}\right)^T \left(z_T - z_{T,s}^E\right) \\ & s.t.\ \boldsymbol{I}_T \boldsymbol{z}_T \leqslant \boldsymbol{J}_T \\ & (\boldsymbol{G}_T)^T \boldsymbol{\omega}_T = \boldsymbol{d}_T \end{aligned} \tag{20b}$$

And let ($\mathbf{z}_{T,c,s}$, $\boldsymbol{\omega}_{T,c,s}$) and $\phi SP\ c,s$ be the solution and objective value of the s-th sub-problem, and update the upper bound $UB = \min\{UB, \mathbf{c}^T \mathbf{x}_{T,c} + \sum_{s \in N^s} p_s \phi_{c,s}^{SP} + (\mathbf{p} - 1)^T \beta_{T,c} + \theta_T \gamma_{T,c}\}$.

Step T-3: If $(UB\text{-}LB)/\mathrm{LB} < \mu_T$, present the results and terminate the procedure; otherwise, update $Z = Z \cup \{\mathbf{z}_{T,c,s}\}$ and $c = c + 1$, and return to Step T-1.

Since problem (17b)-(17c) is linear and always feasible, the CCG algorithm could be terminated in a finite number of iterations [41].

The procedure of the modified nested CCG algorithm for DS sub-problems (which is expressed as Step D-0 to Step D-3) is similar to the procedure above and is detailed in the Appendix I. (SEE Table 1).

4.3. Modified Three-Level cascade ATC algorithm

A modified three-level cascade ATC algorithm is proposed in this part to solve the MPDRRM and precise the dispatch of UAVs. The original MPDRRM can be decomposed into an information sub-problem and a physical sub-problem. The physical sub-problem can be further decomposed into a TS sub-problem and n DS sub-problems. The structure of several DS sub-problems is the same and can be solved in parallel. We iteratively optimize the information and physical sub-problems (7), (13)–(14) by exchanging the operation state of branches and power loads, and optimize the physical sub-problems (13)–(14) by exchanging the transmitted power in the contact lines.

Based on the OCL for UAV dispatching and the CCG for TS and DS sub-problems, the procedure of the modified three-level ATC can be expressed as follows.

Step A-1: Specify the initial values of branch states, coordinates of UAVs, virtual power loads and generations, objective functions of (7), (13)–(14), and penalty function multipliers at each period. Set the iteration counter $w = 0$. Set the convergence tolerances ζ_1, ζ_2, ζ_3.

Step A-2: Start the loop.

Step A-3: Set $w = w + 1$.

Step A-4: Calculate the EDFs in (21a).

$$\widetilde{\varphi}_{t,w-1}(z,x) = \begin{cases} -\lambda_z P_z^D \sum_{\forall x \in \Psi(z)} \widetilde{p}_{x,w-1}^{CBR}(t), \forall z \in \Gamma^{BS} \\ -\widetilde{p}_{z,w-1}^{CBR}(t) P_z^T, \forall z \in \Gamma^{CBR} \\ 0, \forall z \in \Theta^H, z \notin \Gamma^{BS} \cup \Gamma^{CBR} \end{cases}, \forall t \geqslant T^{SS} \tag{21a}$$

$$\widetilde{p}_{z,w-1}^{CBR}(t) = p_{z,T^{SS}}^{CBR} - x_{z,t,w-1}, \forall z \in \Gamma^{CBR}, t \geqslant T^{SS}$$

$$\widetilde{\varphi}_{t,w}^{P}(z) = \widetilde{\varphi}_{t,w}^{I}(z) = \widetilde{\varphi}_{t,w-1}, \forall z \in \Gamma^{CBR}, t \geqslant T^{SS}$$

Step A-5: Conduct Step O-1 to Step O-9 to solve the following problem (21) at each period.

$$\begin{aligned} \min_{P_{t,w}} \widetilde{H}_{V,w}(P_{t,w}, t) = & \widetilde{H}_{V,w}^{T}(P_{t,w}, t) + \sum_{\forall l \in \mathcal{N}_n} \widetilde{H}_{V,w}^{D_l}(P_{t,w}, t) \\ & + \begin{bmatrix} + \left[\kappa_{n+1,t,w}^{U} \left(Z_{t,w}^{IT} - \widetilde{Z}_{t,w}^{PT} \right) + \zeta_{n+1,t,w}^{U} \left(Z_{t,w}^{IT} - \widetilde{Z}_{t,w}^{PT} \right)^2 \right] \\ + \sum_{\forall l \in \mathcal{N}_n} \left[\kappa_{l,t,w}^{U} \left(Z_{t,w}^{ID_l} - \widetilde{Z}_{t,w}^{PD_l} \right) + \zeta_{l,t,w}^{U} \left(Z_{t,w}^{ID_l} - \widetilde{Z}_{t,w}^{PD_l} \right)^2 \right] \end{bmatrix}, \forall t \geqslant T^{SS} \\ Z_{t,w}^{IT} = \sum_{j \in \mathcal{N}_{UT}} \sum_{z \in \widetilde{W}_{j,t,w}} \varphi_{t,w}^{I}(z), & \widetilde{Z}_{t,w}^{PT} = \sum_{j \in \mathcal{N}_{UT}} \sum_{z \in \widetilde{W}_{j,t,w}} \widetilde{\varphi}_{t,w}^{P}(z), \forall t \geqslant T^{SS} \\ Z_{t,w}^{ID_l} = \sum_{j \in \mathcal{N}_{UD_l}} \sum_{z \in \widetilde{W}_{j,t,w}} \varphi_{t,w}^{I}(z), & \widetilde{Z}_{t,w}^{PD_l} = \sum_{j \in \mathcal{N}_{UD_l}} \sum_{z \in \widetilde{W}_{j,t,w}} \widetilde{\varphi}_{t,w}^{P}(z), \forall l \in \mathcal{N}_n, t \geqslant T^{SS} \\ & s.t.\ z \in \Theta^H, \sigma_{j,t,w} \in \Theta^H, \forall j \in \mathcal{N}_U, t \geqslant T^{SS} \\ Vor_{j,t,w} = \left\{ z \in \Theta^H \middle| \|z - \sigma_{i,t,w}\|_2^2 \geqslant \|z - \sigma_{j,t,w}\|_2^2, \forall i \in \mathcal{N}_U, i \neq j \right\}, & \forall j \in \mathcal{N}_U, t \geqslant T^{SS} \\ \widetilde{W}_{j,t,w} = \left\{ z \in \Theta^H \middle| \|z - \sigma_{j,t,w}\|_2^2 \leqslant R_j^{max} \right\}, & \forall j \in \mathcal{N}_U, t \geqslant T^{SS} \end{aligned} \tag{21b}$$

let $P_{t,w} = \{\sigma_{j,t,w} | \forall j \in \mathcal{N}_U\}$, $\forall t \geq T^{SS}$ be the solution of (21) in the w-th loop. Get the sets $\widetilde{W}_{j,t,w}$, $\widetilde{W}_{j,t,w}^{CBR}$, $\widetilde{W}_{j,t,w}^{BS}$ ($\forall j \in \mathcal{N}_U$, $\forall t \geq T^{SS}$) that will be utilized physical sub-problem. Let F_w^U be the objective value of (21).

Step A-6: Solve several DS sub-problems by taking Step D-0 to Step D-3 in parallel, and the l-th DS sub-problem in the w-th loop can be expressed as (22).

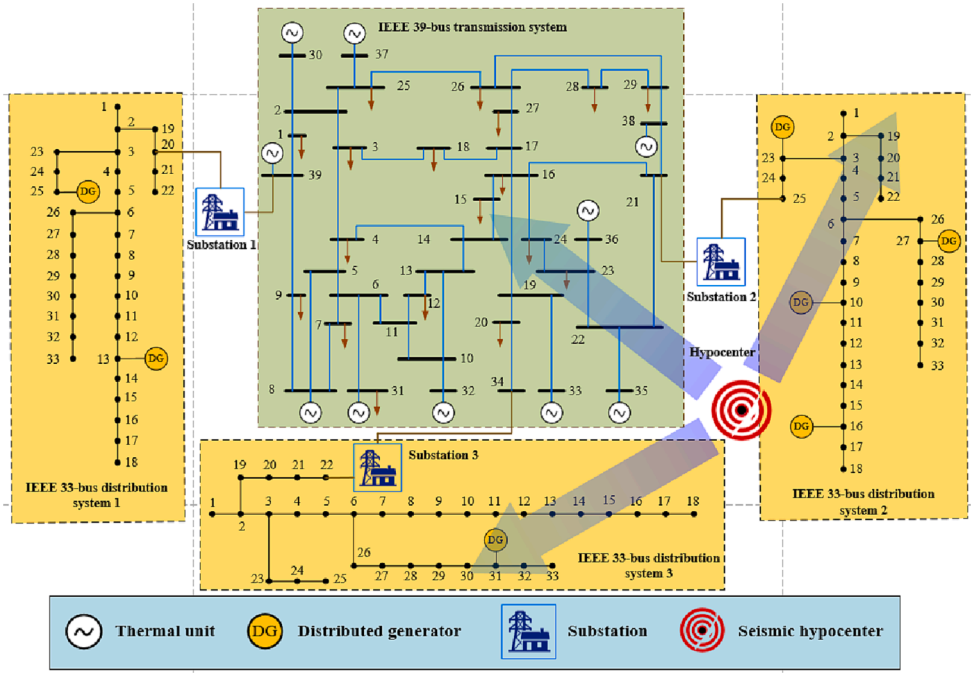


Fig. 6. The topology of T39D99 TDCS.

$$\min_{\boldsymbol{x}_{\mathrm{D}_l,w},\rho_{\mathrm{D}_l,s,w},\boldsymbol{\beta}_{\mathrm{D}_l,w}\geqslant 0,\gamma_{\mathrm{D}_l,w}\geqslant 0,\boldsymbol{\varepsilon}_{\mathrm{D}_l,s,w}} \sum_{\forall s\in N^{\mathrm{S}}} p_s\rho_{\mathrm{D}_l,s,w} + (\boldsymbol{p}_{\mathrm{D}_l}-\mathbf{1})^T\boldsymbol{\beta}_{\mathrm{D}_l,w} + \theta_{\mathrm{D}_l}\gamma_{\mathrm{D}_l,w}$$
$$s.t.\ \boldsymbol{A}_{\mathrm{D}_l}\boldsymbol{x}_{\mathrm{D}_l,w}\leqslant \boldsymbol{b}_{\mathrm{D}_l}$$
$$\left\|\boldsymbol{\varepsilon}_{\mathrm{D}_l,s,w}\right\|_2^2\leqslant\gamma_{\mathrm{D}_l,w}$$
$$\rho_{\mathrm{D}_l,s,w}\geqslant \max_{\boldsymbol{z}_{\mathrm{D}_l,w}\in\Omega}\left[Q_{\mathrm{D}_l,w}\left(\boldsymbol{x}_{\mathrm{D}_l,w},\boldsymbol{z}_{\mathrm{D}_l,w}\right)+\left(\boldsymbol{\beta}_{\mathrm{D}_l,w}\right)^T\boldsymbol{z}_{\mathrm{D}_l,w}-\left(\boldsymbol{\varepsilon}_{\mathrm{D}_l,s,w}\right)^T\left(\boldsymbol{z}_{\mathrm{D}_l,w}-\boldsymbol{z}^{\mathrm{E}}_{\mathrm{D}_l,w}\right)\right]$$
$$\forall l\in\mathcal{N}_{\mathrm{n}} \tag{22a}$$

where

$$Q_{\mathrm{D}_l,w}\left(\boldsymbol{x}_{\mathrm{D}_l,w},\boldsymbol{z}_{\mathrm{D}_l,w}\right)=\min_{\boldsymbol{y}_{\mathrm{D}_l,w}}\left(\boldsymbol{d}_{\mathrm{D}_l}\right)^T\boldsymbol{y}_{\mathrm{D}_l,w} \tag{22b}$$

$$s.t.\ \boldsymbol{E}_{\mathrm{D}_l}\boldsymbol{x}_{\mathrm{D}_l,w}+\boldsymbol{G}_{\mathrm{D}_l}\boldsymbol{y}_{\mathrm{D}_l,w}+\boldsymbol{M}_{\mathrm{D}_l}\boldsymbol{z}_{\mathrm{D}_l,w}\leqslant\boldsymbol{h}_{\mathrm{D}_l}\quad\forall l\in\mathcal{N}_{\mathrm{n}} \tag{22c}$$

Let $\widetilde{\boldsymbol{\varpi}}_{w,t}$ be the solution of equivalent power loads at each period in the TS by solving (22). Let $\{x^{\mathrm{D}_l}_{ij,t,w},\ \forall l\in\mathcal{N}_{\mathrm{n}},\forall t\geq T^{\mathrm{SS}}\}$ be the solution of branch operation states in DSs in the w-th loop. Let $F^{\mathrm{D}}_{w,l}$ be the objective value of the l-th DS sub-problem.

Step A-7: Solve the TS sub-problem (23) by taking Step T-0 to Step T-3.

$$\min_{\boldsymbol{x}_{\mathrm{T},w},\rho_{\mathrm{T},s,w},\boldsymbol{\beta}_{\mathrm{T},w}\geqslant 0,\gamma_{\mathrm{T},w}\geqslant 0,\boldsymbol{\varepsilon}_{\mathrm{T},s,w}} \boldsymbol{c}^T\boldsymbol{x}_{\mathrm{T},w}+\sum_{\forall s\in N^{\mathrm{S}}}p_s\rho_{\mathrm{T},s,w}+(\boldsymbol{p}_{\mathrm{T}}-\mathbf{1})^T\boldsymbol{\beta}_{\mathrm{T},w}+\theta_{\mathrm{T}}\gamma_{\mathrm{T},w}$$
$$s.t.\ \boldsymbol{A}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T},w}\leqslant\boldsymbol{b}_{\mathrm{T}}$$
$$\left\|\boldsymbol{\varepsilon}_{\mathrm{T},s,w}\right\|_2^2\leqslant\gamma_{\mathrm{T},w}$$
$$\rho_{\mathrm{T},s,w}\geqslant\max_{\boldsymbol{z}_{\mathrm{T},w}\in\Omega}\left[Q_{\mathrm{T},w}\left(\boldsymbol{x}_{\mathrm{T},w},\boldsymbol{z}_{\mathrm{T},w}\right)+\left(\boldsymbol{\beta}_{\mathrm{T},w}\right)^T\boldsymbol{z}_{\mathrm{T},w}-\left(\boldsymbol{\varepsilon}_{\mathrm{T},s,w}\right)^T\left(\boldsymbol{z}_{\mathrm{T},w}-\boldsymbol{z}^{\mathrm{E}}_{\mathrm{T},s}\right)\right] \tag{23a}$$

where

$$Q_{\mathrm{T},w}\left(\boldsymbol{x}_{\mathrm{T},w},\boldsymbol{z}_{\mathrm{T},w}\right)=\min_{\boldsymbol{y}_{\mathrm{T},w}}(\boldsymbol{d}_{\mathrm{T}})^T\boldsymbol{y}_{\mathrm{T},w} \tag{23b}$$

$$s.t.\ \boldsymbol{E}_{\mathrm{T}}\boldsymbol{x}_{\mathrm{T},w}+\boldsymbol{G}_{\mathrm{T}}\boldsymbol{y}_{\mathrm{T},w}+\boldsymbol{M}_{\mathrm{T}}\boldsymbol{z}_{\mathrm{T},w}\leqslant\boldsymbol{h}_{\mathrm{T}} \tag{23c}$$

Table 2
Data of UAVs in each system.

Number	System	System area (km^2)	MASP	Optimum flight altitude (km)	Maximum radius (km)
4	TS	2039	117	9.4	10.2
3	DS 1	338	111	5.1	4.8
3	DS 2	431	114	6.6	5.6
3	DS 3	406	111	5.1	4.8

Let $\widetilde{\boldsymbol{\vartheta}}_{k,l,t}$ be the solution of equivalent power generations at each period in the l-th DS by solving (23). Let $\{x^{\mathrm{T}}_{ij,t,w},\ \forall t\geq T^{\mathrm{SS}}\}$ be the solution of branch operation states in the TS. Let F^{T}_{w} be the objective value of (23).

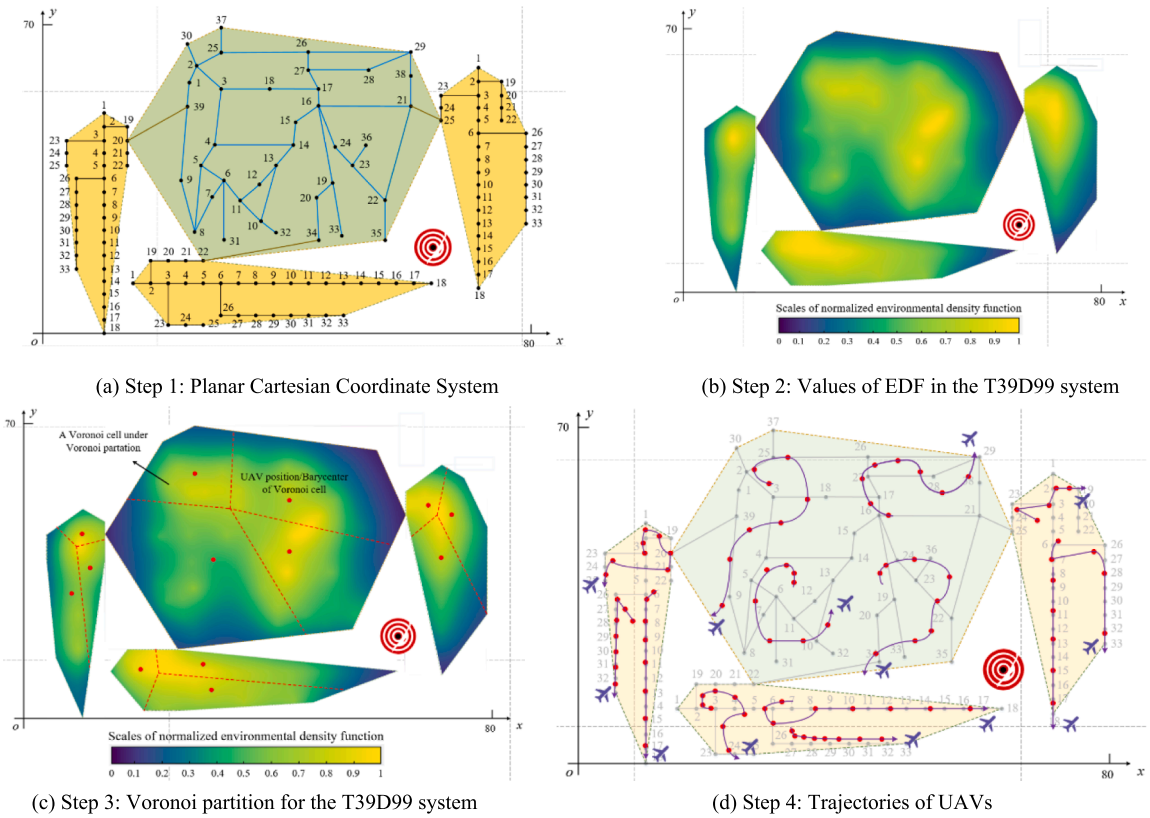
Step A-8: Calculate the EDFs in (24).

$$\varphi^{\mathrm{P}}_{t,w}(z)=\widetilde{\varphi}_{t,w}(z,x)=\begin{cases}-\lambda_z P^{\mathrm{D}}_z\sum\limits_{\forall x\in\Psi(z)}\widetilde{p}^{\mathrm{CBR}}_{x,w}(t),\forall z\in\Gamma^{\mathrm{BS}}\\ -\widetilde{p}^{\mathrm{CBR}}_{z,w}(t)P^{\mathrm{T}}_z,\forall z\in\Gamma^{\mathrm{CBR}}\\ 0,\forall z\in\Theta^{\mathrm{H}},z\notin\Gamma^{\mathrm{BS}}\cup\Gamma^{\mathrm{CBR}}\end{cases} \tag{24}$$
$$\widetilde{p}^{\mathrm{CBR}}_{z,w}(t)=p^{\mathrm{CBR}}_{z,T^{\mathrm{SS}}}-x_{z,t,w},\forall z\in\Gamma^{\mathrm{CBR}},t\geqslant T^{\mathrm{SS}}$$
$$\varphi^{\mathrm{I}}_{t,w}(z)=\widetilde{\varphi}^{\mathrm{P}}_{t,w}(z),\forall z\in\Gamma^{\mathrm{CBR}},t\geqslant T^{\mathrm{SS}}$$

Step A-9: Check the following convergence condition (24)

$$\begin{cases}\sum\limits_{\forall t\geqslant T^{\mathrm{SS}}}\left\|\mathrm{Z}^{\mathrm{PT}}_t-\mathrm{Z}^{\mathrm{IT}}_t\right\|_2^2+\sum\limits_{\forall l\in\mathcal{N}_{\mathrm{n}}}\left\|\mathrm{Z}^{\mathrm{PD}_l}_t-\mathrm{Z}^{\mathrm{ID}_l}_t\right\|_2^2\leqslant\zeta^{\mathrm{L}}_1\\ \sum\limits_{\forall t\geqslant T^{\mathrm{SS}}}\left\|\widetilde{\boldsymbol{\varpi}}_{w,t}-\widetilde{\boldsymbol{\vartheta}}_{w,l}\right\|_2^2\leqslant\zeta^{\mathrm{L}}_2\\ \sum\limits_{\forall t\geqslant T^{\mathrm{SS}}}\dfrac{\left(F^{\mathrm{T}}_w+\sum\limits_{\forall l\in\mathcal{N}_{\mathrm{n}}}F^{\mathrm{D}}_{w,l}+F^{\mathrm{U}}_w\right)-\left(F^{\mathrm{T}}_{w-1,l}+\sum\limits_{\forall l\in\mathcal{N}_{\mathrm{n}}}F^{\mathrm{D}}_{w-1,l}+F^{\mathrm{U}}_{w-1,l}\right)}{F^{\mathrm{T}}_{w,l}+\sum\limits_{\forall l\in\mathcal{N}_{\mathrm{n}}}F^{\mathrm{D}}_{w,l}+F^{\mathrm{U}}_{w,l}}\leqslant\zeta^{\mathrm{L}}_3\end{cases} \tag{25}$$

If the result satisfies (25), stop the loop, let $\{\mathbf{x}_w,\ \mathbf{y}_w,\ \mathbf{z}_w,\ \mathrm{P}_w\}$ be the



(a) Step 1: Planar Cartesian Coordinate System

(b) Step 2: Values of EDF in the T39D99 system

(c) Step 3: Voronoi partition for the T39D99 system

(d) Step 4: Trajectories of UAVs

Fig. 7. The procedure for generating the trajectory of UAVs in the T39D99 system.

optimal results of the MPDRRM; otherwise, update the penalty function multipliers by using (26), and turn to Step A-3.

$$\begin{cases} \kappa_{l,t}^{\mathrm{P},w} = \kappa_{l,t}^{\mathrm{P},w-1} + 2\left(\varsigma_{l,t}^{\mathrm{P},w-1}\right)^2\left(P_{l,t,w}^{\mathrm{T}\to\mathrm{D}} - P_{l,t,w}^{\mathrm{D}\to\mathrm{T}}\right) \\ \kappa_{l,t}^{\mathrm{Q},w} = \kappa_{l,t}^{\mathrm{Q},w-1} + 2\left(\varsigma_{l,t}^{\mathrm{Q},w-1}\right)^2\left(Q_{l,t,w}^{\mathrm{T}\to\mathrm{D}} - Q_{l,t,w}^{\mathrm{D}\to\mathrm{T}}\right) \\ \kappa_{l,t}^{\mathrm{U},w} = \kappa_{l,t}^{\mathrm{U},w-1} + 2\left(\varsigma_{l,t}^{\mathrm{U},w-1}\right)^2\left(\mathrm{Z}_{t,w}^{\mathrm{PD}_l} - \mathrm{Z}_{t,w}^{\mathrm{ID}_l}\right) \\ \kappa_{n+1,t}^{\mathrm{U},w} = \kappa_{n+1,t}^{\mathrm{U},w-1} + 2\left(\varsigma_{n+1,t}^{\mathrm{U},w-1}\right)^2\left(\mathrm{Z}_{t,w}^{\mathrm{PT}} - \mathrm{Z}_{t,w}^{\mathrm{IT}}\right) \\ \varsigma_{l,t}^{\mathrm{P},w} = \delta^{\mathrm{U}}\varsigma_{l,t}^{\mathrm{P},w-1} \\ \varsigma_{l,t}^{\mathrm{Q},w} = \delta^{\mathrm{U}}\varsigma_{l,t}^{\mathrm{Q},w-1} \\ \varsigma_{l,t}^{\mathrm{U},w} = \delta^{\mathrm{U}}\varsigma_{l,t}^{\mathrm{U},w-1} \\ 2 \leqslant \delta^{\mathrm{U}} \leqslant 3 \end{cases} \quad (26)$$

$$\forall l \in \mathcal{N}_{\mathrm{n}}, \mathrm{t} \geqslant \mathrm{T}^{\mathrm{SS}}$$

5. Results and discussion

In this section, we use a T39D99 TDCS, which contains a modified IEEE 39-bus TS [55] and three modified IEEE 33-bus DSs [45], to verify the validity and effectiveness of the proposed model. The related model is solved by GUROBI 8.1 on a personal computer with an Intel Core i7-9700 CPU and 16 GB RAM.

5.1. Case setting

The topology of T39D99 TDCS is shown in Fig. 6, including 5 DGs in the DSs and 10 TUs in the TS. The data of TUs and DGs are available in Appendix II. 3 UAVs are utilized for each DS, while 4 UAVs are utilized in the TS. According to [42], the parameters of UAVs for each system are set in Table 2. In the TS, the MASP is relatively higher than that in the DSs, leading to a much higher flight altitude and larger coverage zone of UAVs than those in the DSs. The situation of high signal loss in DSs also leads to a necessity to apply more UAVs to support communication. Since the signal propagation environments of these three DSs are different and incurred by the man-made structures and inner scattering [42–43], the MASP of DS1 or DS 3 is relatively lower than that of DS 2, resulting in various flight altitudes and maximum radiuses in three DSs. As for MSs, we assume that each DS would install 3 MSs dispatching in the system to support power loads.

5.2. Results of the UAV dispatching

In this paper, UAVs are applied in the third stage to recover the information system and cooperate with system restoration, the dispatching procedures of which are shown in Fig. 7. To implement our proposed model, a Planar Cartesian Coordinate System together with the convex hulls of the TS and DSs covering the T39D99 system are first constructed in Fig. 7(a). A typical earthquake [50] is assumed to generate around the TS, DS2, and DS3. Then, based on the seismic moment, the outage probability of branches, and the distribution of power loads, we formulate the EDF for the T39D99, the values of which are shown in Fig. 7(b) as a heat map. In Fig. 7(b), the EDF takes higher values close to the power load center, contact lines, and seismic hypocenter. The Voronoi partition can be configured according to the distribution of the EDF. By applying Step O-1 to Step O-9, we can calculate each Voronoi cell of the T39D99 system for one period, and the optimal UAV positions can also be accessed, which is shown in Fig. 7(c). Finally, the optimal dispatch of UAVs coordinated with the system restoration at each period is designed according to the proposed modified three-level ATC algorithm, the trajectories of which are shown in Fig. 7(d).

Table 3
Reinforcement result for the T39D99.

System	Hardened branches			Adaptive MS		Reinforcement cost (10^3 $)		
	Case 1	Case2	Case 3	Case 1	Case 3	Case 1	Case 2	Case 3
TS	21–22, 21–38, 29–38, 20–34, 19–33, 16–19, 16–21	21–22, 21–38, 23–36, 20–34, 19–33, 22–35, 16–21	21–22, 21–38, 22–35, 22–35, 23–36, 19–33, 16–19	—	—	3305.1	3302.2	3323.2
DS 1	21–22	21–22	21–22	Renting MSs with maximum output of 103.7 kW for one MS; total capacity 4902.2 kWh of all MS	Renting MSs with maximum output of 332.3 kW for one MS; total capacity 7921.4 kWh of all MS	369.3	158.2	530.6
DS 2	7–8, 8–9, 9–10, 3–23, 15–16, 26–27	6–26, 8–9, 9–10, 3–23, 15–16, 26–27	7–8, 8–9, 9–10, 3–23, 6–26, 26–27	Renting MS with 0 kW, 0 kWh	Renting MS with 0 kW, 0 kWh	2393.2	2386.3	2387.5
DS 3	9–10, 8–9, 16–17, 17–18	9–10, 8–9, 16–17, 17–18	9–10, 30–31, 8–9, 31–32	Renting MSs with maximum output of 321.3 kW for one MS; total capacity 6956.1 kWh of all MS	Renting MSs with maximum output of 774.1 kW for one MS; total capacity 13936.5 kWh of all MS	1421.9	522.1	2493.2

Table 4
Restoration results for DS 2.

System	Case 1		Case 2		Case 3	
	Power restoration (10^3 MW)	Operation cost (10^3 $)	Power restoration (10^3 MW)	Operation cost (10^3 $)	Power restoration (10^3 MW)	Operation cost (10^3 $)
TS	2.731	5132.7	2.681	5302.5	2.367	5543.2
DS 1	0.762	399.3	0.724	425.6	0.692	465.3
DS 2	0.157	28.2	0.156	28.2	0.156	28.3
DS 3	1.245	1938.2	1.166	2019.7	1.083	2293.7
Total	4.895	7498.4	4.727	7776.1	4.298	8330.5

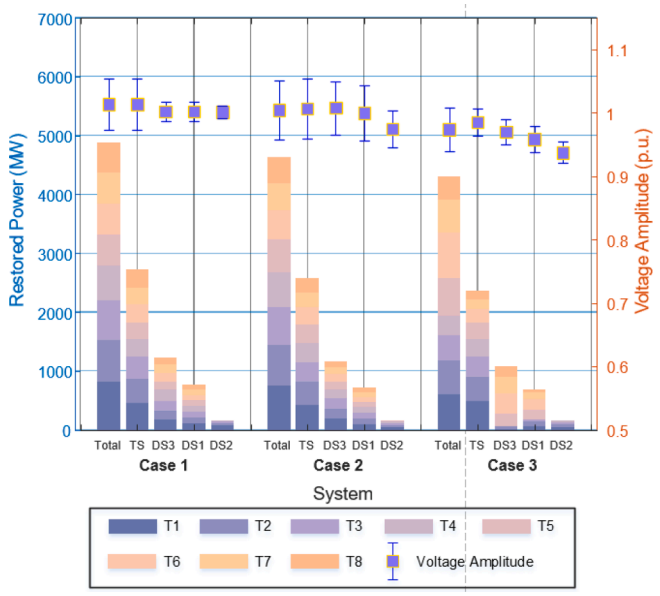


Fig. 8. Power restoration and voltage amplitude.

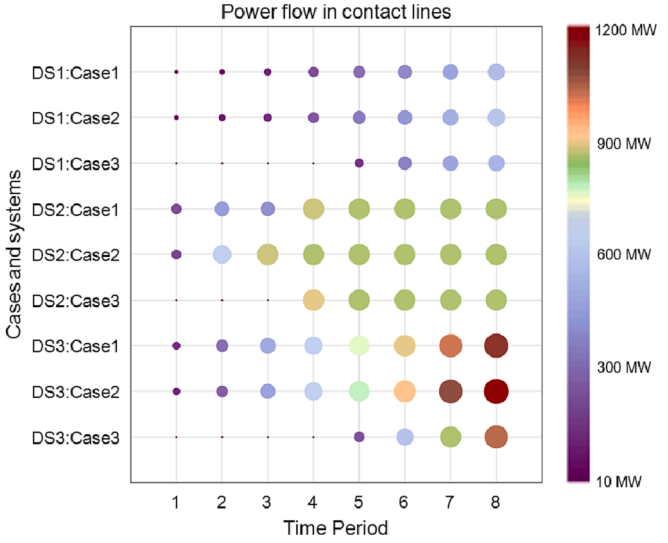
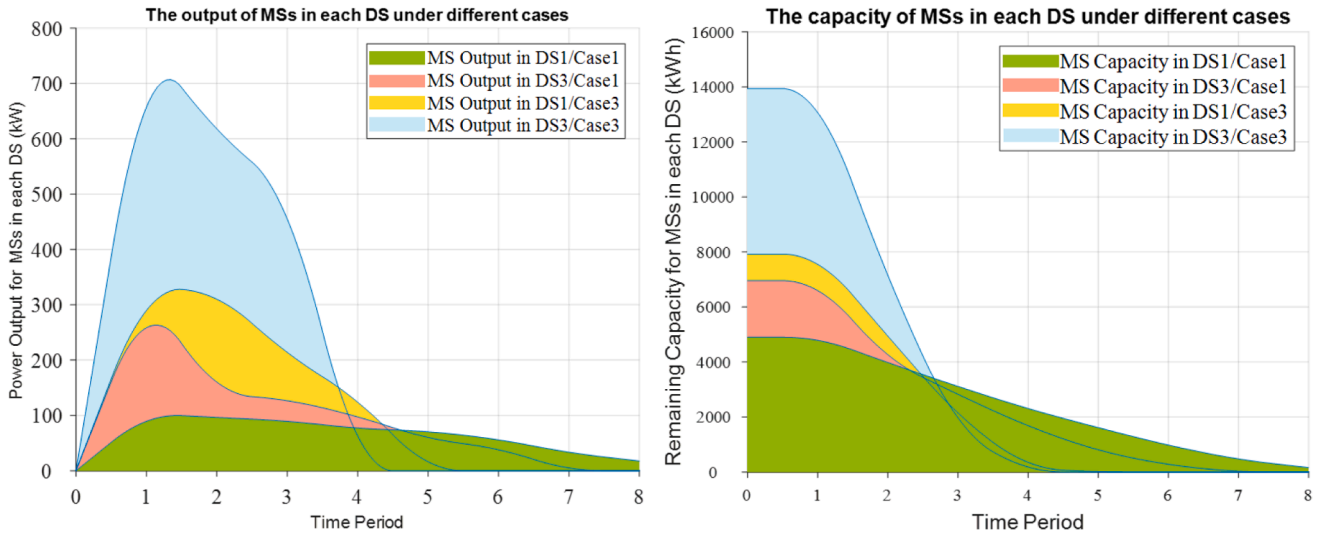


Fig. 9. Power flow in contact lines.

5.3. Results and benefits of MS and UAVs

We use three cases to investigate the benefits of TS-DS collaboration and dispatch of UAVs and MS. In Case 1, UAVs are applied, and the TS and DSs are coordinately reinforced and operated. In Case 2, the application of MS is not included. In Case 3, MS is rented in the TS and DSs, but UAVs are not dispatched to recover the communication system. The resilience strategy in Case 3 for TS and DSs is optimized separately due to failures in the communication system. For each DS, the DSO first utilizes DGs to restore critical loads. Other power loads are restored by the TS after the system is re-connected to the main network and the communication between several systems is recovered.

The reinforcement results of three cases for the modified T39D99 TDCS are shown in Table 3. The reinforcement strategy is implemented when early warnings of the earthquake are received. Several branches are hardened and MS is rented before the occurrence of the earthquake. The branches around the seismic hypocenter and connected to the generators are first hardened to ensure power supply under the



(a) The output of MSs in each DS under different cases

(b) The capacity of MSs in each DS under different cases

Fig. 10. The capacity and output of MSs during the restoration process.

Table 5
The restoration results and computation times of three models.

Number of faulted branches	MPDRRM		ATC-NUM		ADMM-NUM	
	Power restoration (10^3 MW)	Computation time (s)	Power restoration (10^3 MW)	Computation time (s)	Power restoration (10^3 MW)	Computation time (s)
10	0.432	720	0.416	542	0.421	694
20	1.274	2310	0.988	1492	1.025	1762
25	1.923	4902	1.656	3603	1.773	3996
30	2.433	7939	1.992	5021	2.175	6034

Table A1
Data of TUs and DGs in T39D99 TDCS.

Type	Bus	Units	System	Max output (MW)	Min output (MW)	Ramp (MW/min)	Offering price ($/MWh)
DG	13, 25	3	DS 1	30 × 3	6 × 3	2	31.273
DG	10, 16, 23, 27	5	DS 2	50 × 5	10 × 5	2.67	28.926
DG	31	2	DS 3	30 × 2	6 × 2	1.33	33.274
TU	30, 36	1	TS	600 × 1	150 × 1	10	18.919
TU	32, 35, 37	2	TS	350 × 2	100 × 2	7	23.667
TU	31, 33, 34	1	TS	800 × 1	200 × 1	12	15.345
TU	38	3	TS	250 × 3	80 × 3	5	27.924
TU	39	2	TS	500 × 2	150 × 2	10	18.919

earthquake. The branches around the contact lines (i.e., branches 29–38, 20–34, 16–21 in the TS) are also hardened to ensure power transmission between several systems in Case 1 and Case 2.

As for the MS, the rented MS in Case 1 has a lower capacity and power output than those in Case 3. The total power output and capacity of rented MS in Case 1 are 46.9 % and 45.7 % lower than those in Case 3. As shown in Table 3, the output of DGs can cover the power output in DS 2, leading to a non-adaption of MS in the system. The plenty output of DGs in DS 2 also results in a reversed supply from DS 2 to TS when the earthquake occurrence, which can decrease the reinforcement cost and operation cost. The less application of MS also leads to lower MS rental costs. Thus, the total reinforcement cost in Case 1 is 14.3 % lower than that in Case 3.

In the third stage, by applying the proposed MPDRRM, the restoration results containing the power restorations, operation costs, and voltage fluctuations are shown in Table 4, Fig. 8.

The total power restoration in Case 1 is 13.89 % higher than that in Case 3, indicating that the power interaction between several systems can increase the efficiency of load restoration. In Case 1, UAVs can recover the communication system between the TS and DSs, which make it available for the exact transfer of energy requirements in different systems. The power generated in DS 2 can be immediately supplied to TS, leading to a 3.55 % higher power restoration of TS than that in Case 3. For the same reason, the power restorations in DS 1 and DS 3 are respectively 4.62 % and 7.66 % higher than those in Case 3. Note that the capacity of DGs can cover power loads in DS 2. The power restorations and operation costs are nearly the same in these three cases.

As for Case 2, the total power restoration in Case 2 is 3.43 % lower than that in Case 1, indicating that some critical loads in Case 2 cannot be supported promptly by MS once the earthquake occurs. The higher power loss at critical loads also leads to a higher penalty cost in Case 2 and the total operation cost is 3.70 % higher than that in Case 1.

The restoration at each period and voltage amplitude variations for each system are shown in Fig. 8, and the power flow in contact lines is shown in Fig. 9. Compared with Case 1 and Case 2, the power restorations in Case 3 during the first three periods are much lower, and over 40 % power restorations are concentrated in the last three periods. In

addition, the power flow in the contact lines during the first three periods is much lower and surges to over 300 MW in the 5th period, indicating that DSs are reconnected to the main grid and supported by the TS after the 4th period in Case 3. On the contrary, the power restorations in Case 1 and Case 2 are more evenly distributed in 8 periods, and the deviation of power restorations is around 27.2 %, while that in Case 3 is around 39.6 %. The power load restoration in Case 3 leads to a large fluctuation range of voltage amplitude, and the fluctuation range of voltage amplitude in Case 3 is 0.136p.u., which is 38.2 % and 62.5 % higher than that in Case 1 and Case 2.

The capacity and output of MSs during the restoration process are also investment in this section. Fig. 10 illustrates the changes in MSs' capacity and output during the restoration process. Since the capacity of DGs in the 2nd DS is higher than the power loads, no MS is rented in the 2nd DS. And in Case 2, we assume that there is no MS in the TDCS. So, Fig. 10 only shows the change of MS output and capacity in the 1st DS and the 2nd DS under Case 1 and Case 3. As the earthquake occurs, the output power of all MSs begins to increase and reaches the maximum output before the 2nd time period. In Fig. 9, we can access that three DSs and TS are coordinated together in Case 1 so that the power support to the 1st DS and the 3rd DS from the TS and the 3rd DS is higher than that in Case 3. Therefore, it can be observed that the output of MSs in Case 3 is much higher throughout the whole restoration process, and the stored energy of MSs is fully consumed before the 5th time period. And after the 5th time period, the power loads in the 1st DS and the 3rd DS are mainly supported by the TS so that the power output of MSs in every DS under two cases gradually reduces. Besides, since the capacity of DGs in the 1st DS is higher than that in the 3rd DS, the maximum output and the installed capacity of MSs in the 1st DS are lower than those in the 3rd DS. For the same reason, the output of MSs in the 3rd DS is higher than that in the 1st DS before the 5th time period. So, the stored energy of MSs in the 3rd DS is consumed much faster than that in the 1st DS.

5.4. Performance of the proposed method

In this section, we compare our method with traditional methods proposed in references [16] and [22]. The authors in [16] proposed a coordinated restoration model of the transmission and distribution system with a decentralized scheme, and the ATC algorithm was applied to solve this model. In [22], the authors proposed a similar restoration model and used the ADMM method to solve it. Also, the model in [22] considered the benefit of DGs while restoration. In these two models, UAVs and MSs were not considered. For such a situation, we construct two models with different algorithms: the ATC-based TDCS restoration model without UAVs and MSs (ATC-NUM), and the ADMM-based TDCS restoration model without UAVs and MSs (ADMM-NUM). All models are verified in the T39D99 TDCS. The power restoration results and computation times of our model (MPDRRM), ATC-NUM, and ADMM-NUM with different faulted branches are shown in Table 5.

When there are fewer faulted lines, the amount of load loss is relatively low, and fewer power loads are restored by these three models. At this time, due to the lower amount of load shedding, the restored power loads of these three models are relatively similar. When there are 10 faulted branches, since the DGs are taken into account in MPDRRM and ADMM-NUM, the restored power loads in these two models are 3.84 % and 1.21 % higher than those in ATC-NUM. With the increasing number of faulted branches, the load shedding progressively gets worse. Under such conditions, since MPDRRM considers the dispatch of MSs and UAVs, the gap between the restored power loads in these three models gradually widened. When there are 30 faulted branches, the restored power loads in MPDRRM is 2.433 GW, which is respectively 22.13 % and 11.86 % higher than that in ATC-NUM and ADMM-NUM. And since ADMM-NUM considers the dispatch of DG, the restored power loads in ADMM-NUM are consistently higher than that in ATC-NUM.

As for the computation times, the computation times of MPDRRM are consistently higher than those in the other two models because more binary variables about UAVs and MSs are contained in MPDRRM. When the number of faulted branches is relatively low, the continuous variables and binary variables in these three models are closer. But when the number of faulted branches keeps increasing, the number and capacity of UAVs and MSs also increase so that the gap between the computation times of MPDRRM and the other two models is widening. And when there are 30 faulted branches, the computation time of MPDRRM is 7939s, which is respectively 58.12 % and 31.57 % higher than that in ATC-NUM and ADMM-NUM.

6. Conclusion

In this paper, we propose an MPDRRM for the TDCS. The failure in the communication system is also taken into consideration. The MS is applied and dispatched during the restoration to support critical loads and maintain voltage stability. To address the unavailability of the communication system during the network restoration, UAVs are applied to construct a temporary communication system while restoring. The optimal dispatching of UAVs is modeled as a multi-period coverage function optimization problem, in which the OCL algorithm is utilized to solve this problem. Then, a modified three-level ATC algorithm embedded with a nested CCG method and OCL algorithm is used to solve the proposed MPDRRM. Numerical tests in the T39D99 show that the proposed model can conspicuously improve the resilience of the TDCS against natural disasters. By applying MS before the occurrence of the earthquake, the proposed model has a 15.7 % and 10 % reduction in the reinforcement cost and the operation cost. The lower operation cost indicates that MS can flexibly support many critical loads before they are restored by DGs or TUs. The application of UAVs and MS can also improve power load restoration and the fluctuation of voltage amplitude.

CRediT authorship contribution statement

Zekai Wang: Writing – original draft. **Tao Ding:** Conceptualization. **Chenggang Mu:** Software. **Yuhan Huang:** Methodology. **Miao Yang:** . **Yueyang Yang:** Investigation. **Yi Lin:** . **Meng Li:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix I:. Procedure of the modified nested CCG for the distribution system sub-problems

Similar to that of the TS sub-problem, the procedure of the modified nested CCG for the distribution system sub-problem can be detailed as follows.

Step D-0: Specify an initial branch outage set $Z_T = \{\mathbf{z}_{D,0,s}\}$ and a convergence tolerance μ_D. Set the lower bound LB to -∞, the upper bound UB to +∞, and the iteration counter $c = 1$.

Step D-1: Solve the following master problem.

$$\begin{aligned}
&\min_{\boldsymbol{x}_D,\rho_{D,s},\boldsymbol{\beta}_D\geqslant 0,\gamma_D\geqslant 0,\boldsymbol{\varepsilon}_{D,s}} \sum_s p_s\rho_{D,s} + (\boldsymbol{p}_D - \mathbf{1})^T\boldsymbol{\beta}_D \ + \ \theta_D\gamma_D \\
&\qquad s.t.\ \boldsymbol{A}_D\boldsymbol{x}_D \leqslant \boldsymbol{b}_D \\
&\qquad \left\|\boldsymbol{\varepsilon}_{D,s}\right\|_2^2 \leqslant \gamma_D \\
&\rho_{D,s} \geqslant (\boldsymbol{d}_D)^T\boldsymbol{y}_{D,l,s} + (\boldsymbol{\beta}_D)^T\boldsymbol{z}_{D,l,s} - \left(\boldsymbol{\varepsilon}_{D,s}\right)^T\left(\boldsymbol{z}_{D,l,s} - \boldsymbol{z}_{D,s}^E\right), \forall l < c \\
&\qquad \boldsymbol{E}_D\boldsymbol{x}_D + \boldsymbol{G}_D\boldsymbol{y}_{D,l,s} + \boldsymbol{M}_D\boldsymbol{z}_{D,l,s} \leqslant \boldsymbol{h}_D, \forall l < c
\end{aligned} \tag{AI-1-a}$$

Then, let ($\mathbf{x}_{D,c}$, $\rho_{D,c,s}$, $\boldsymbol{\beta}_{D,c}$, $\gamma_{D,c}$, $\boldsymbol{\varepsilon}_{D,c,s}$) and ϕMP c be the solution and objective value of the master problem, and update the lower bound $LB = \phi$MP c.

Step D-2: Solve the corresponding N^s sub-problems based on the optimal solution of the master problem derived in Step D-1. The s-th sub-problem is given as follows.

$$\begin{aligned}
&\max_{\boldsymbol{\omega}_D\leqslant 0}(\boldsymbol{h}_D - \boldsymbol{E}_D\boldsymbol{x}_D - \boldsymbol{M}_D\boldsymbol{z}_D)^T\boldsymbol{\omega}_D \ + \ \left(\boldsymbol{\beta}_{D,c}\right)^T\boldsymbol{z}_D - \left(\boldsymbol{\varepsilon}_{D,c,s}\right)^T\left(\boldsymbol{z}_D - \boldsymbol{z}_{D,s}^E\right) \\
&\qquad s.t.\ \boldsymbol{I}_D\boldsymbol{z}_D \leqslant \boldsymbol{J}_D \\
&\qquad (\boldsymbol{G}_D)^T\boldsymbol{\omega}_D = \boldsymbol{d}_D
\end{aligned} \tag{AI-1-b}$$

And let ($\mathbf{z}_{D,c,s}$, $\boldsymbol{\omega}_{D,c,s}$) and ϕSP c,s be the solution and objective value of the s-th sub-problem, and update the upper bound $UB = \min\{UB, \sum_{s\in N^s} p_s\phi_{c,s}^{SP} + (\mathbf{p} - 1)^T\beta_{D,c} + \theta_D\gamma_{D,c}\}$.

Step D-3: If (UB-LB)/LB < μ_D, present the results and terminate the procedure; otherwise, update $Z = Z\cup\{\mathbf{z}_{D,c,s}\}$ and $c = c + 1$, and return to Step D-1.

Appendix II. : Data of TUs and DGs in the T39D99 TDCS

The data of TUs and DGs are available in Table A1. The maximum power load of T39D99 TDCS is 6.25×10^3 MW. Among the three DSs, DS 1 and DS 3 are regarded as two urban systems with massive critical power customers, while DS 2 is a sub-urban system with high capacities of DSs.

References

[1] Zhao N, Yu X, Hou K, et al. Full-time scale resilience enhancement framework for power transmission system under ice disasters. Int J Electr Power Energy Syst 2021;126(106609).

[2] Galvan E, Mandal P, Sang Y, et al. Networked microgrids with roof-top solar PV and battery energy storage to improve distribution grids resilience to natural disasters. Int J Electr Power Energy Syst 2020;123(106239).

[3] Shi Q, Liu W, Zeng B, et al. Enhancing distribution system resilience against extreme weather events: Concept review, algorithm summary, and future vision. Int J Electr Power Energy Syst 2022;138(107860).

[4] Shi Q, Li F, Olama M, et al. Network reconfiguration and distributed energy resource scheduling for improved distribution system resilience. Int J Electr Power Energy Syst 2020;124(106355).

[5] Wang Z, Ding T, Jia W, et al. Multi-stage stochastic programming for resilient integrated electricity and natural gas distribution systems against typhoon natural disaster attacks. Renew Sust Energ Rev 2022;159(111784).

[6] Khan I, Ghassemi M. A probabilistic approach for analysis of line outage risk caused by wildfires. Int J Electr Power Energy Syst 2022;139(108042).

[7] Vazquez D, Qiu F, Fan N, et al. Wildfire Mitigation Plans in Power Systems: A Literature Review. IEEE Trans Power Syst 2022;37(5):3540–51.

[8] Nazemi M, Dehghanian P. Powering Through Wildfires: An Integrated Solution for Enhanced Safety and Resilience in Power Grids. IEEE Trans Ind Appl 2022;58(3):4192–202.

[9] Zhao Q, Du Y, Zhang T, et al. Resilience index system and comprehensive assessment method for distribution network considering multi-energy coordination. Int J Electr Power Energy Syst 2021;133(107211).

[10] Xie H, Sun X, Chen C, et al. Resilience Metrics for Integrated Power and Natural Gas Systems. IEEE Trans Smart Grid 2022;13(3):2483–6.

[11] Kandaperumal G, Pandey S, Srivastava A, et al. AWR: Anticipate, Withstand, and Recover Resilience Metric for Operational and Planning Decision Support in Electric Distribution System. IEEE Trans Smart Grid 2022;13(1):179–90.

[12] Panteli M, Mancarella P, Trakas DN, et al. Metrics and Quantification of Operational and Infrastructure Resilience in Power Systems. IEEE Trans Power Syst 2017;32(6):4732–42.

[13] Panteli M, Trakas DN, Mancarella P, et al. Power Systems Resilience Assessment: Hardening and Smart Operational Enhancement Strategies. Proc of the IEEE 2017;105(7):1202–13.

[14] Gautam P, Piya P, Karki R. Resilience Assessment of Distribution Systems Integrated With Distributed Energy Resources. IEEE Trans Sustain Energy 2021;13(1):338–48.

[15] Wang Q, Lin S, Yang Y, et al. A decomposition and coordination algorithm for SVSM interval of integrated transmission and distribution networks considering the uncertainty of renewable energy. Int J Electr Power Energy Syst 2022;136(107761).

[16] Zhao J, Wang H, Liu Y, et al. Coordinated Restoration of Transmission and Distribution System Using Decentralized Scheme. IEEE Trans Power Syst 2019;34(5):3428–42.

[17] Kargarian A, Mehrtash M, Falahati B, et al. Decentralized implementation of unit commitment with analytical target cascading: a parallel approach. IEEE Trans Power Syst 2018;33(4):3981–93.

[18] Shen Z, Liu M, Xu L, et al. Coordinated scheduling of integrated transmission and distribution systems using an improved Lipschitz dynamic programming approach. Int J Electr Power Energy Syst 2022;140(108076).

[19] Liu J, Tang Z, Zeng P, et al. Distributed adaptive expansion approach for transmission and distribution networks incorporating source-contingency-load uncertainties. Int J Electr Power Energy Syst 2022;136(107711).

[20] Jin X, Mu Y, Jia H, et al. Alleviation of overloads in transmission network: A multi-level framework using the capability from active distribution network. Int J Electr Power Energy Syst 2019;112:232–51.

[21] Rad H, Moravej Z. An approach for simultaneous distribution, sub-transmission, and transmission networks expansion planning. Int J Electr Power Energy Syst 2017;91:166–82.

[22] Roofegari Nejad R, Sun W, Golshani A. Distributed Restoration for Integrated Transmission and Distribution Systems With DERs. IEEE Trans Power Syst 2019;34(6):4964–73.

[23] Saha S, Aldeen M. Generalised scalable fault dependent time invariant state space model for large interconnected power systems. Int J Electr Power Energy Syst 2014;60:165–81.

[24] Zhao J, Liu Y, Wang H, et al. Receding horizon load restoration for coupled transmission and distribution system considering load-source uncertainty. Int J Electr Power Energy Syst 2020;116(105517).

[25] Bastami H, Shakarami M, Doostizadeh M. A non-hierarchical ATC framework for parallel scheduling of active distribution network with multiple autonomous microgrids. Int J Electr Power Energy Syst 2021;133(107293).

[26] Liao W, Li P, Wu Q, et al. Distributed optimal active and reactive power control for wind farms based on ADMM. Int J Electr Power Energy Syst 2021;129(106799).

[27] Rajaei A, Fattaheian-Dehkordi S, Fotuhi-Firuzabad M, et al. Decentralized transactive energy management of multi-microgrid distribution systems based on ADMM. Int J Electr Power Energy Syst 2021;132(107126).

[28] Mansouri S, Ahmarinejad A, Ansarian M, et al. Stochastic planning and operation of energy hubs considering demand response programs using Benders decomposition approach. Int J Electr Power Energy Syst 2020;120(106030).

[29] Zhao J, Zhang Z, Yao J, et al. A distributed optimal reactive power flow for global transmission and distribution network. Int J Electr Power Energy Syst 2019;104:524–36.

[30] Oghaddam A, et al. A decentralized robust model for optimal operation of distribution companies with private microgrids. Int J Electr Power Energy Syst 2019;106:105–23.

[31] Li Z, Guo Q, Sun H, et al. Coordinated economic dispatch of coupled transmission and distribution systems using heterogeneous decomposition. IEEE Trans Power Syst 2016;31(6):4817–30.
[32] Ji X, Zhang Y, Han X, et al. Multi-level interactive unit commitment of regional power system. Int J Electr Power Energy Syst 2021;125(106464).
[33] Zhao J, Wang W, Guo C. Hierarchical optimal configuration of multi-energy microgrids system considering energy management in electricity market environment. Int J Electr Power Energy Syst 2023;144(108572).
[34] Xin S, Guo Q, Sun H, et al. Cyber-physical modeling and cyber-contingency assessment of hierarchical control systems. IEEE Trans Smart Grid 2015;6(5): 2375–85.
[35] Ti B, Li G, Zhou M, et al. Resilience assessment and improvement for cyber-physical power systems under typhoon disasters. IEEE Trans Smart Grid 2022;13(1): 783–94.
[36] Feng Y, Hu B, Hao H, et al. Design of Distributed Cyber-Physical Systems for Connected and Automated Vehicles With Implementing Methodologies. IEEE Trans Ind Informat 2018;14(9):4200–11.
[37] Qin Z, Mo Y, Liu H, et al. Operational flexibility enhancements using mobile energy storage in day-ahead electricity market by game-theoretic approach. Energy 2021; 232.
[38] Abdeltawab H, Mohamed Y. Mobile energy storage scheduling and operation in active distribution systems. IEEE Trans Ind Electron 2017;64:6828–40.
[39] https://baijiahao.baidu.com/s?id=1705961097108803459&wfr=spider&for=pc.
[40] Zhou Y, Wei Z, Shahidehpour M, et al. Distributionally robust resilient operation of integrated energy systems using moment and wasserstein metric for contingencies. IEEE Trans Power Syst 2021;36(4):3574–84.
[41] Chen C, Shen X, Guo Q, et al. Adaptive robust planning-operation co-optimization of energy hubs with modified column-and-constraint generation algorithm. IEEE Power Energy Soc Gener Meet (PESGM) 2019;2019:1–5.
[42] Al-Hourani A, Kandeepan S, Lardner S. Optimal LAP altitude for maximum coverage. IEEE Wireless Commun Lett 2014;3(6):569–72.
[43] Cortes J, Martinez S, Karatas T, Bullo F. Coverage control for mobile sensing networks. IEEE Trans Rob Auto 2004;20(2):243–55.
[44] Jadbabaie A, Lin J, Morse AS. Coordination of groups of mobileautonomous agents using nearest neighbor rules. IEEE Trans Automat Contr 2003;48:988–1001.
[45] Okabe A, Boots B, Sugihara K, et al. "Spatial Tessellations: Concepts and Applications of Voronoi Diagrams. 2nd ed,. New York: Wiley; 2000.
[46] Klein R. Concrete and abstract voronoi diagrams, ser. Lect Notes Comput Sci, vol. 400. New York, NY: Springer-Verlag; 1989.
[47] Du Q, Faber V, Gunzburger M. Centroidal Voronoi tessellations: Applications and algorithms. SIAM Rev 1999;41(4):637–76.
[48] Zollo A, Nazeri S, Colombelli S. Earthquake seismic moment, rupture radius, and stress drop from P-wave displacement amplitude versus time curves. IEEE Trans Geoscience and Remote Sensing 2022;60:1–11.
[49] Nazemi M, Moeini-Aghtaie M, Fotuhi-Firuzabad M, et al. Energy storage planning for enhanced resilience of power distribution networks against earthquakes. IEEE Trans Sustain Enery 2022;11(2):795–806.
[50] Liu Z, Zhang J, Li X, et al. Long-Term resilience curve analysis of wenchuan earthquake-affected counties using DMSP-OLS nighttime light images. IEEE J Selected Top Appl Earth Observ Remote Sens 2021;14:10854–74.
[51] Ma S, Chen B, Wang Z. Resilience enhancement strategy for distribution systems under extreme weather events. IEEE Trans Smart Grid 2018;9(2):1442–51.
[52] Duan C, Fang W, Jiang L, et al. Distributionally robust chance-constrained approximate AC-OPF with Wasserstein metric. IEEE Trans Power Syst 2018;33(5): 4924–36.
[53] Ding T, Wang Z, Jia W, et al. Multiperiod distribution system restoration with routing repair crews, mobile electric vehicles, and soft-open-point networked microgrids. IEEE Trans Smart Grid 2020;11(6):4795–808.
[54] Zeng B, Zhao L. Solving two-stage robust optimization problems using a column-and-constraint generation method. Oper Res Lett 2013;41(5):457–61.
[55] Liu F, Bie Z, Wang X. Day-ahead dispatch of integrated electricity and natural gas system considering reserve scheduling and renewable uncertainties. IEEE Trans Sustain Enery 2019;10(2):646–58.